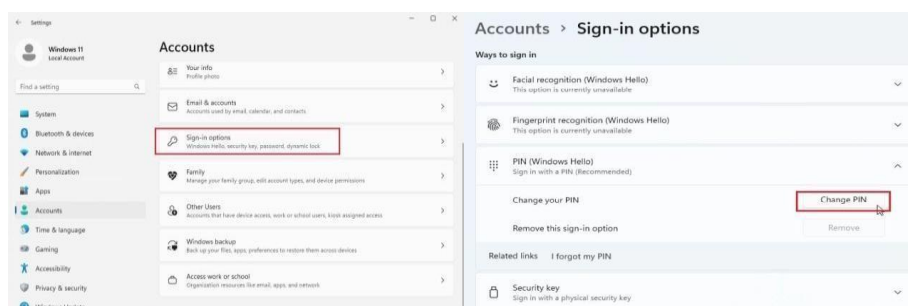


Expt. 01) Change/set Windows desktop security Pin/Password and check Windows Update system**Password setup or change pin**

Step 01: Press the **Windows** keyboard shortcut “**Windows + I**” to open the **Settings** app. Now, move to **Accounts** → **Sign-in** options.



Step 02: Here, click to expand the “**Password**” section and then click the “**Change PIN**” button.



Step 03: After that enter the current password of your Windows PC and click on “**Next**”.

Step 04: On the next page, you can change the password easily. You can also add a hint to help you recover your account in case you forget the password.

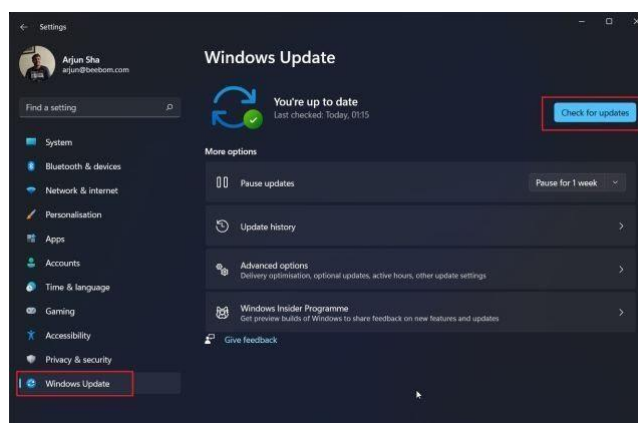


Step 05: Finally click on “**Finish**”, and you are done. You have successfully changed your Windows password.

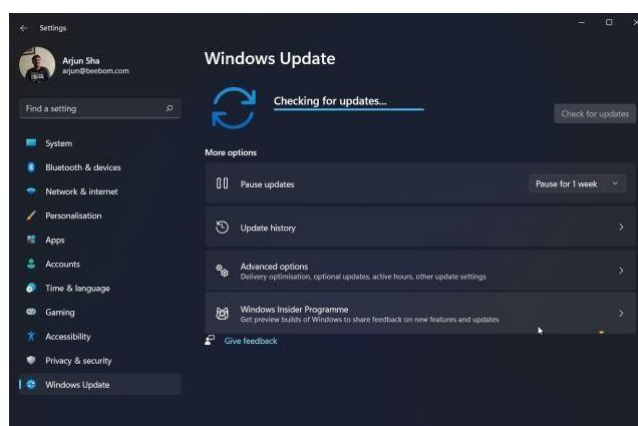


Check for Windows Updates and Update the System

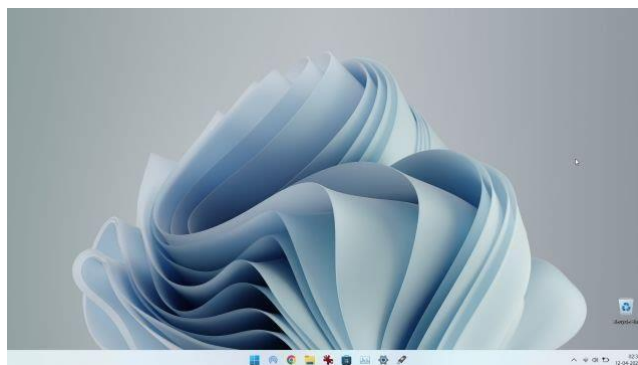
Step 01: First press the shortcut “**Windows + I**” to open the Settings app. Next, navigate to the “**Windows Update**” section from the left sidebar.

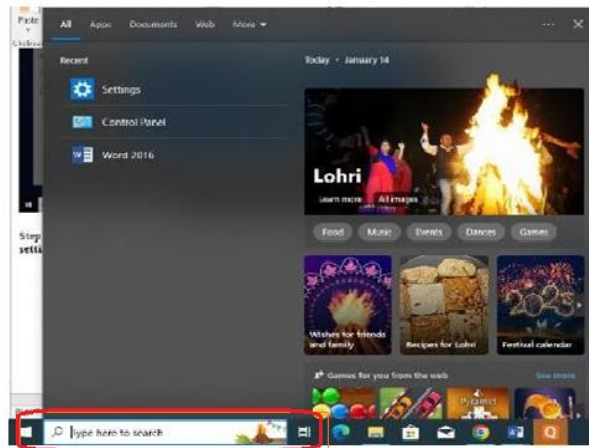
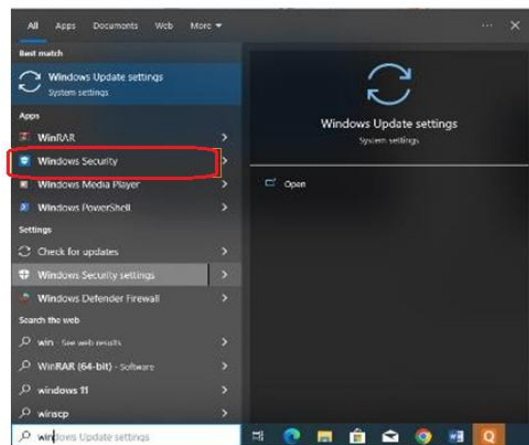
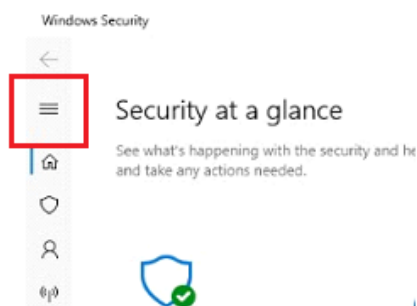


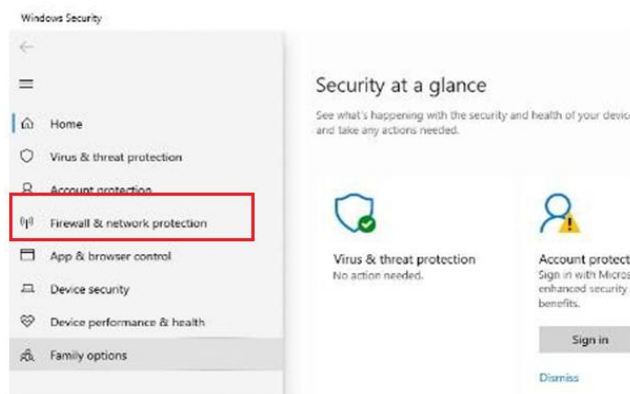
Step 02: Once here, click on “**Check for Updates**”. If there is an update available, it will show up here and will be downloaded automatically.



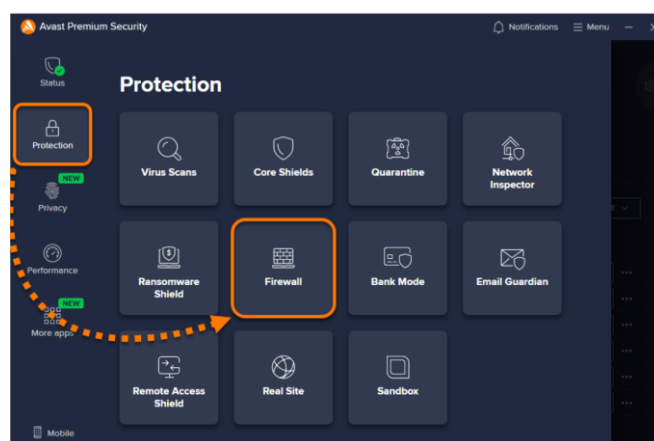
Step 03: After that, the update will be installed, and you will be asked to restart your PC. Simply reboot the computer to enable the recently installed updates.



Expt. 02) Demonstrate Turning on and off Windows OS Firewall**Step 01: Go to Search in Windows taskbar****Step 02: Search Windows Security****Step 03: Click on Navigation button****Step 04: Click on "Firewall and network protection"**



Step 05: If you have installed any anti-virus software, the firewall settings will open in that application. Now you can turn firewall on and off.

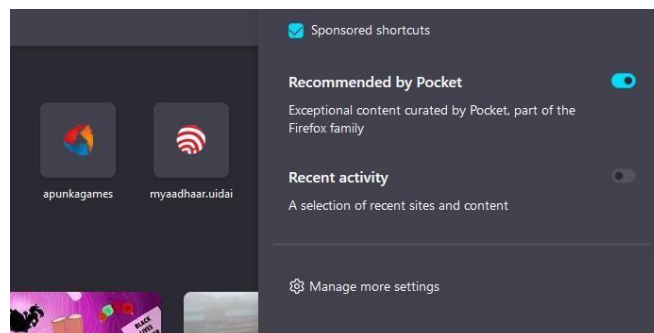


Expt. 03) Check the browser and website certificates and analyze them**Viewing Browser certificates**

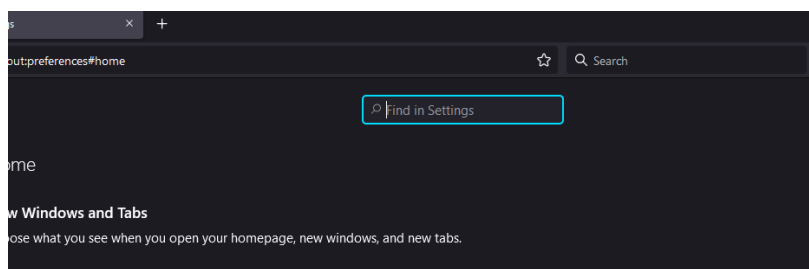
Step 01: Open **Firefox** browser and click on **“Settings”** icon



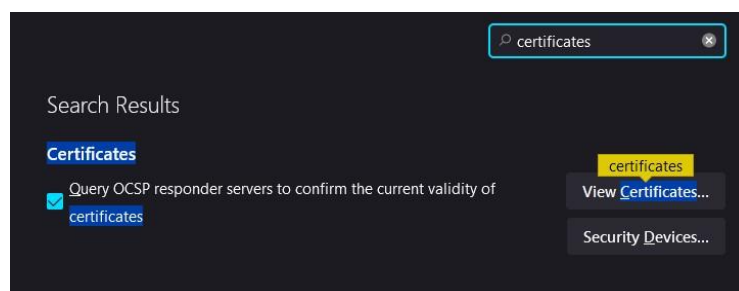
Step 02: Click the **“Manage more settings”** option



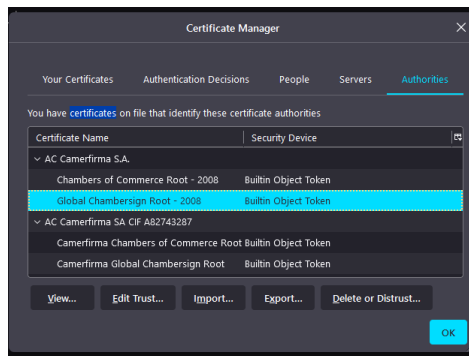
Step 03: Search for **certificates** in the search box



Step 04: Click on **“View Certificates”**



Step 05: Select any one certificate and click **“View”**



Step 06: Now you can see the certificate

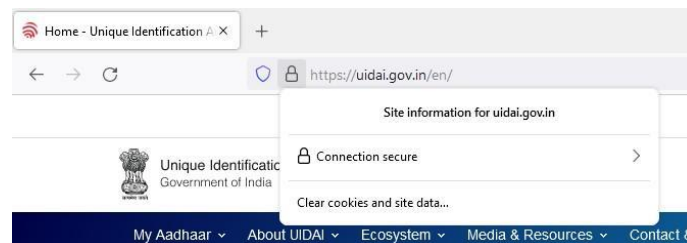
Website certificate check

Step 01: Go to “Firefox” and search any website

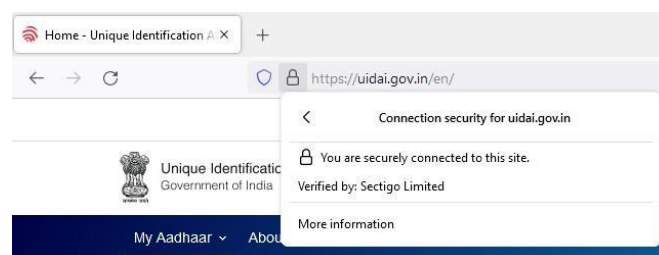


Step 02: Click the **lock** icon in the top-left corner of the address bar

Step 03: Click “Connection secure”

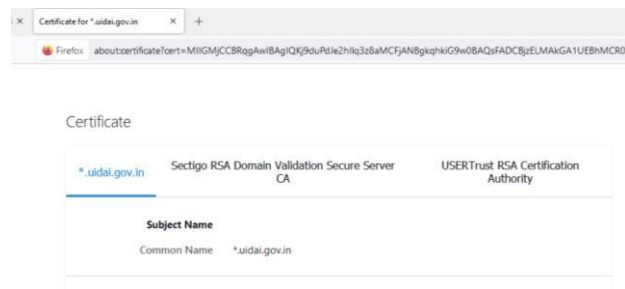


Step 04: Click “More information”



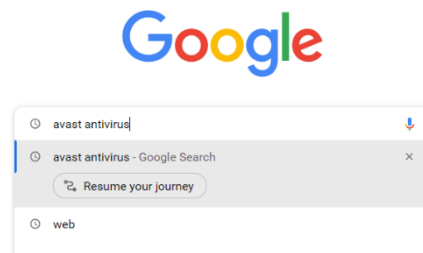
Step 05: Now click view certificate



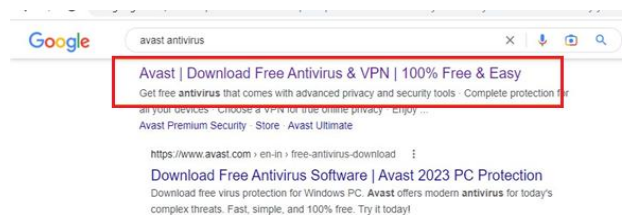
Step 06: Now you can see the certificate details

Expt. 04) Install Antivirus and scan the computer.

Step 01: Open any browser and search **Avast** (or any other) antivirus.



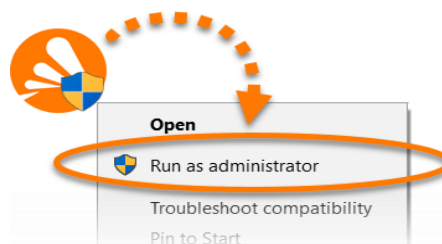
Step 02: click on **Avast | download free antivirus & VPN link**



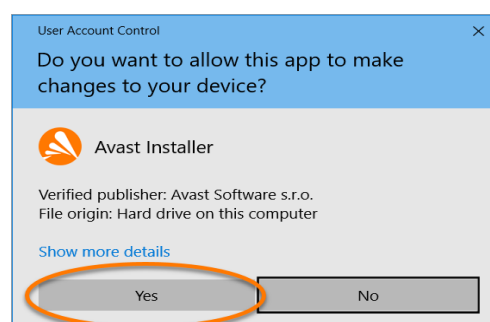
Step 03: Click on **“Download Free Protection”** button



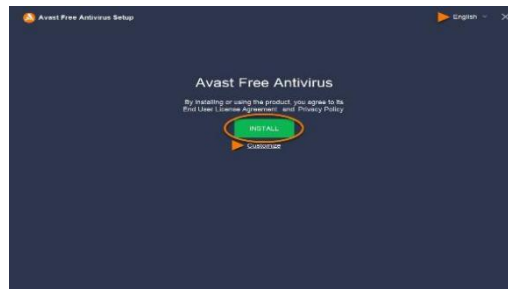
Step 04: Right-click the downloaded setup file and select **“Run as administrator”**



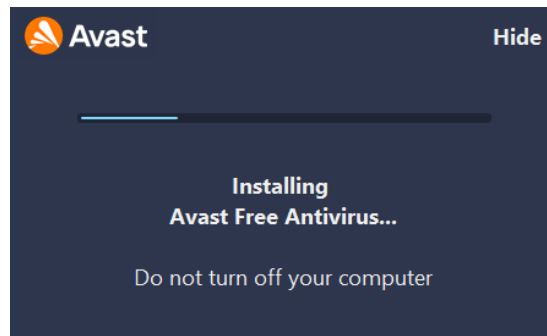
Step 05: If prompted for permission by the **User Account Control** dialog, click **“Yes”**.



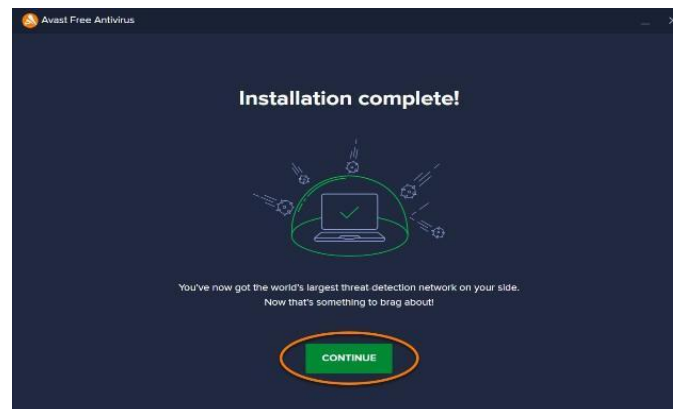
Step 06: Then, click **Install**



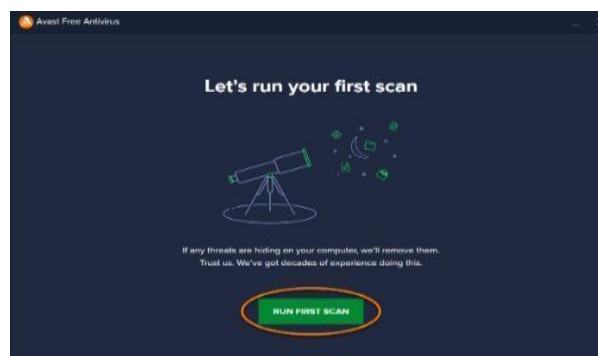
Step 07: Wait while setup installs **Avast Free Antivirus** on your PC.



Step 08: When the installation is complete, click **Continue**.



Step 09: Click **Run first scan** to start a comprehensive **Smart Scan**,



Expt. 05) Install Git application and perform the basic Git operations below

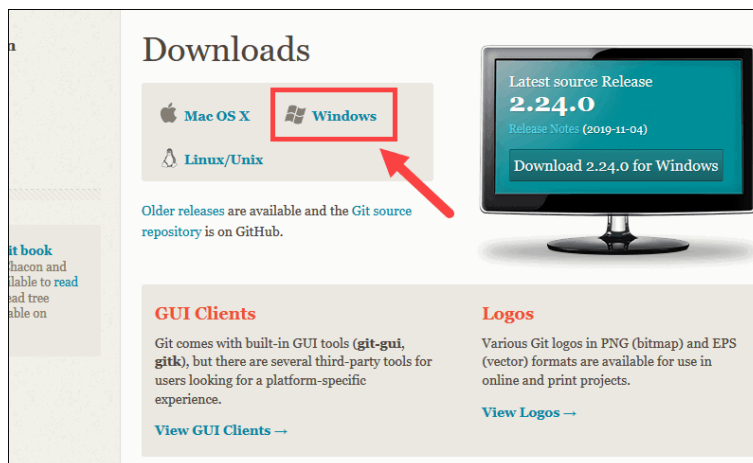
- *Create a repository*
- *Cloning a repository*
- *Making and committing repository changes*
- *Viewing the history of all changes*

Steps for Installing Git for Windows

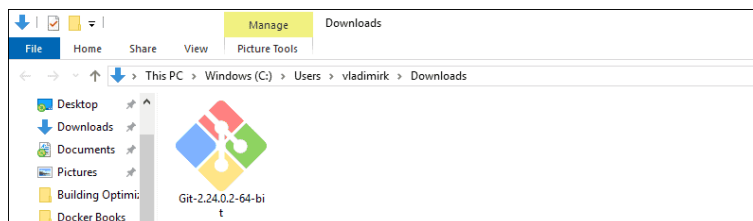
Installing **Git** prompts you to select a text editor. If you don't have one, we strongly advise you to install prior to installing **Git**.

Download Git for Windows

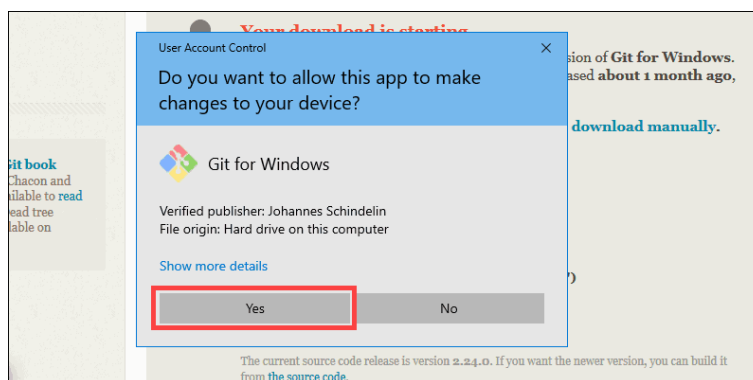
1. Browse to the official Git website: <https://git-scm.com/downloads>
2. Click the download link for Windows and allow the download to complete.

**Extract and Launch Git Installer**

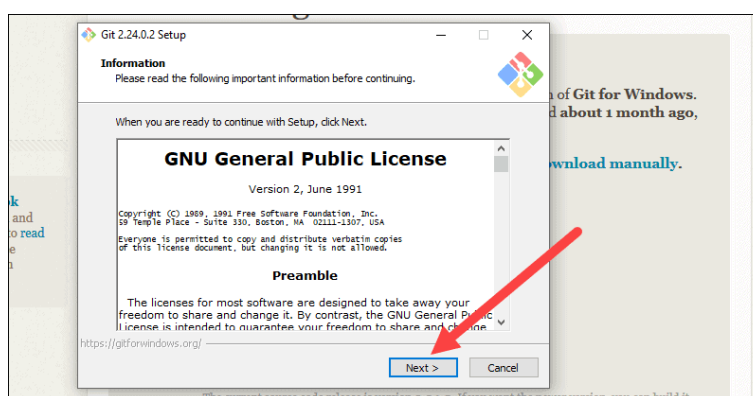
3. Browse to the download location (or use the download shortcut in your browser). Double-click the file to extract and launch the installer.



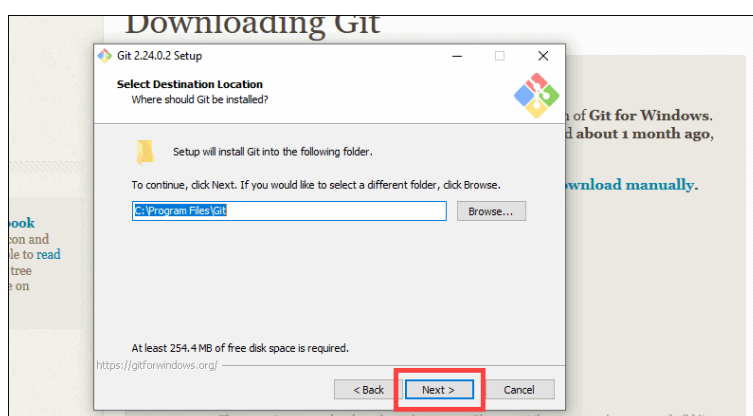
4. Allow the app to make changes to your device by clicking **Yes** on the User Account Control dialog that opens.



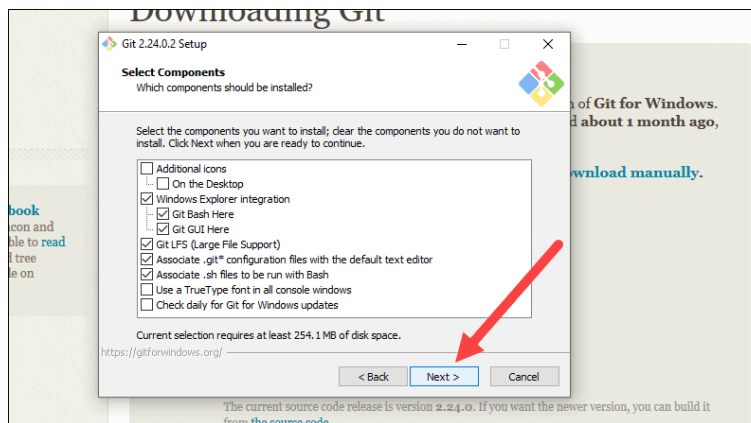
5. Review the [GNU General Public License](#), and when you're ready to install, click **Next**.



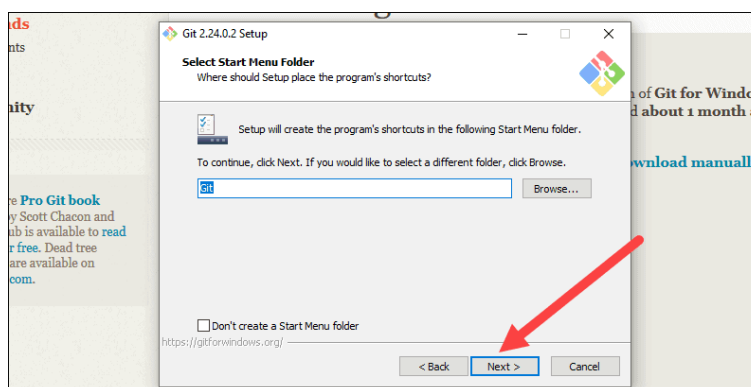
6. The installer will ask you for an installation location. Leave the default, unless you have reason to change it, and click **Next**.



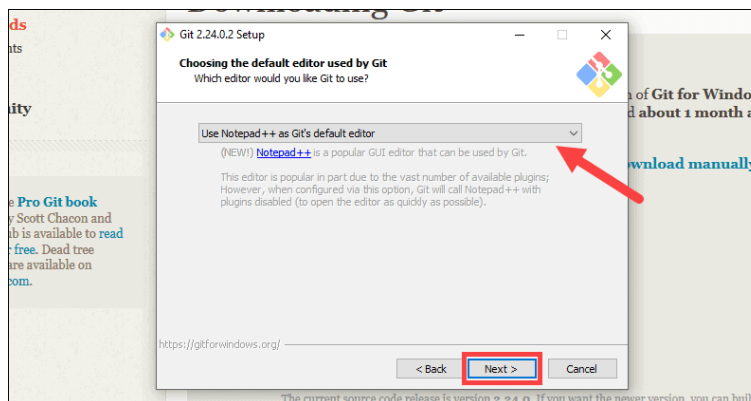
7. A component selection screen will appear. Leave the defaults unless you have a specific need to change them and click **Next**.



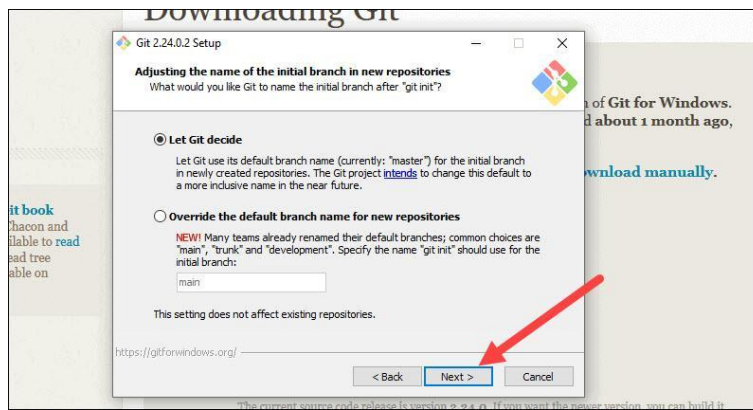
1. The installer will offer to create a Start Menu folder. Simply click **Next**.



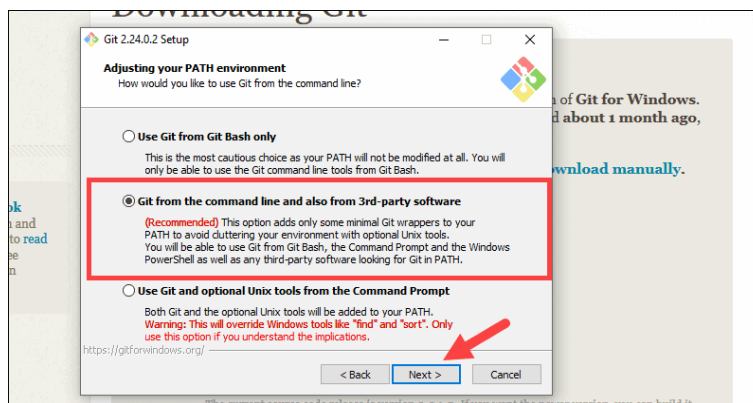
2. Select a text editor you'd like to use with Git. Use the drop-down menu to select Notepad++ (or whichever text editor you prefer) and click **Next**.



3. The next step allows you to choose a different name for your initial branch. The default is '**master**.' Unless you're working in a team that requires a different name, leave the default option and click **Next**.

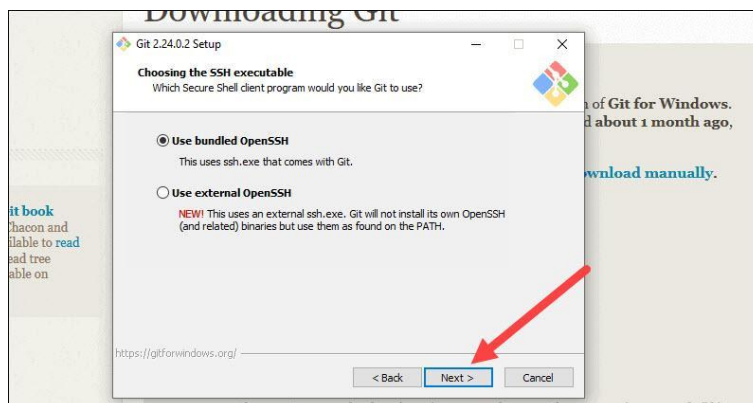


4. This installation step allows you to change the **PATH environment**. The **PATH** is the default set of directories included when you run a command from the command line. Leave this on the middle (recommended) selection and click **Next**.

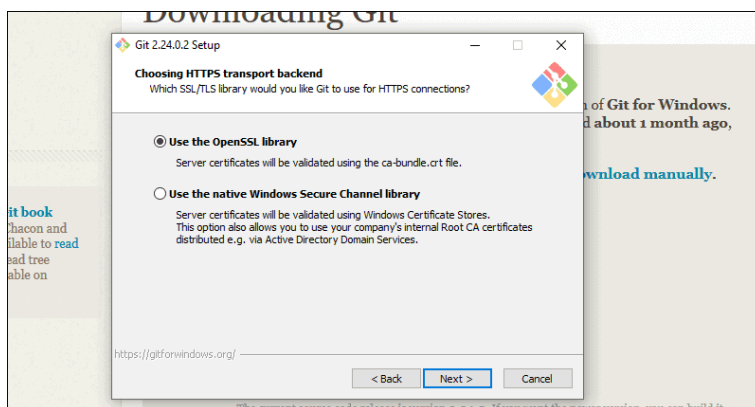


Server Certificates, Line Endings and Terminal Emulators

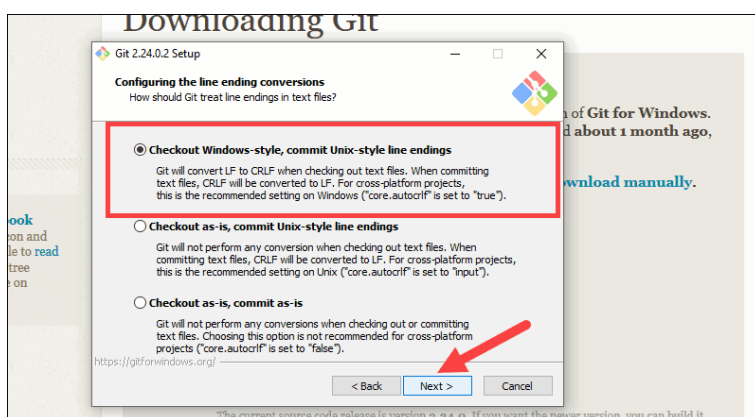
5. The installer now asks which **SSH** client you want **Git** to use. **Git** already comes with its own **SSH** client, so if you don't need a specific one, leave the default option and click **Next**.



6. The next option relates to server certificates. Most users should use the default. If you're working in an Active Directory environment, you may need to switch to Windows Store certificates. Click **Next**.



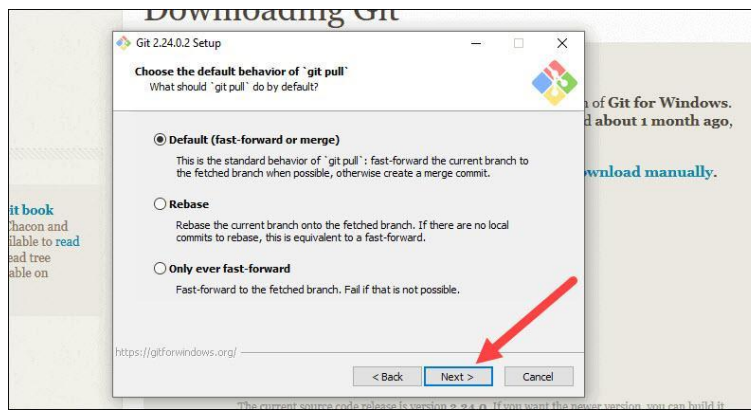
7. The next selection converts line endings. It is recommended that you leave the default selection. This relates to the way data is formatted and changing this option may cause problems. Click **Next**.



8. Choose the [terminal emulator](#) you want to use. The default **MinTTY** is recommended, for its features. Click **Next**.



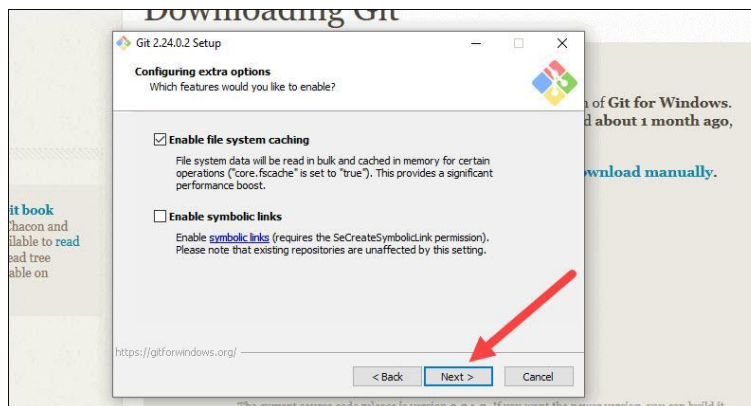
9. The installer now asks what the **git pull** command should do. The default option is recommended unless you specifically need to change its behavior. Click **Next** to continue with the installation.



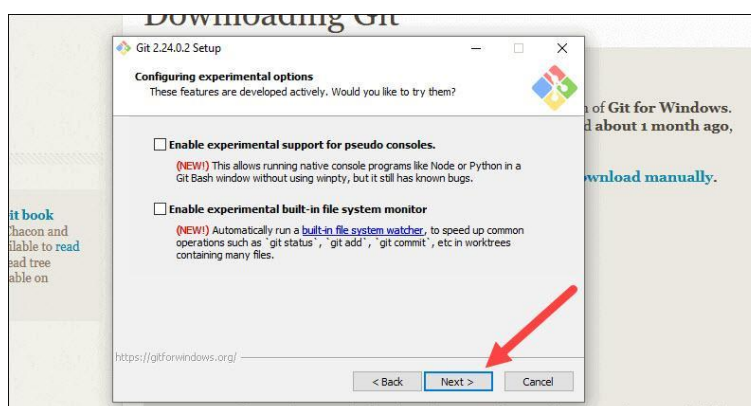
1. Next you should choose which credential helper to use. Git uses credential helpers to fetch or save credentials. Leave the default option as it is the most stable one, and click **Next**.

Additional Customization Options

2. The default options are recommended; however, this step allows you to decide which extra option you would like to enable. If you use symbolic links, which are like shortcuts for the command line, tick the box. Click **Next**.

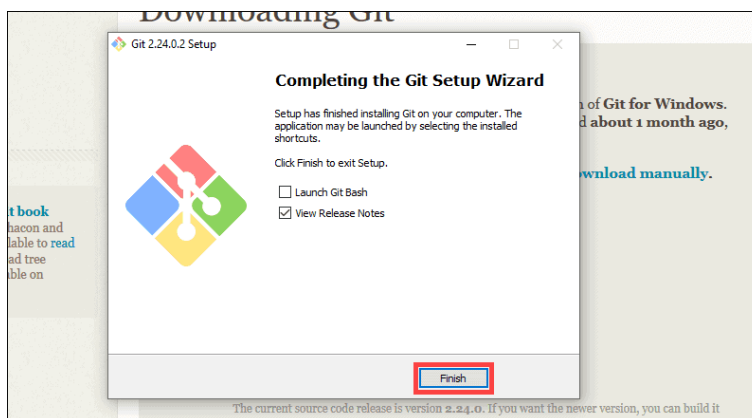


3. Depending on the version of Git you're installing, it may offer to install experimental features. At the time this article was written, the options to include support for pseudo consoles and a built-in file system monitor were offered. Unless you are feeling adventurous, leave them unchecked and click **Install**.



Complete Git Installation Process

4. Once the installation is complete, tick the boxes to view the Release Notes or Launch Git Bash, then click **Finish**.

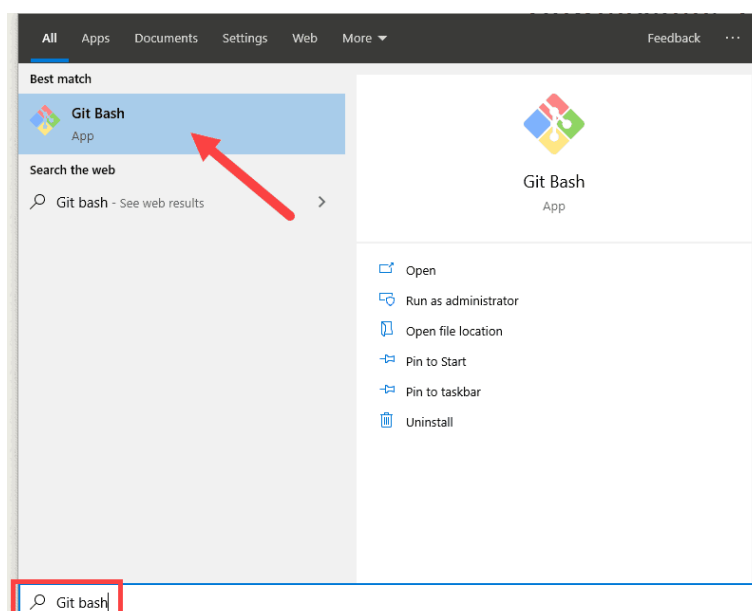


1. How to Launch Git in Windows

Git has two modes of use – a **bash scripting shell** (or command line) and a **graphical user interface (GUI)**.

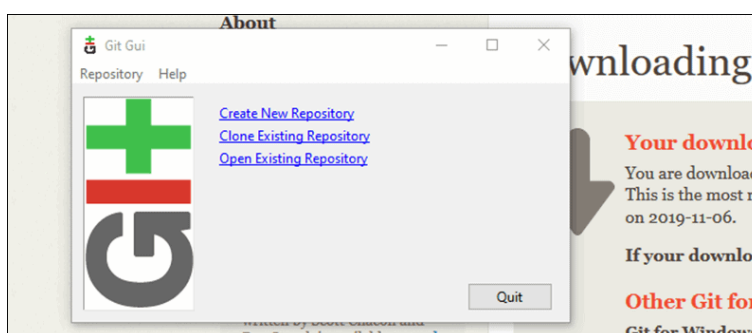
- **Launch Git Bash Shell**

To launch **Git Bash** open the **Windows Start** menu, type **git bash** and press **Enter** (or click the application icon).



- **Launch Git GUI**

To launch **Git GUI** open the **Windows Start** menu, type **git gui** and press **Enter** (or click the application icon).



2. Connecting to a Remote Repository

You need a GitHub username and password for this next step.

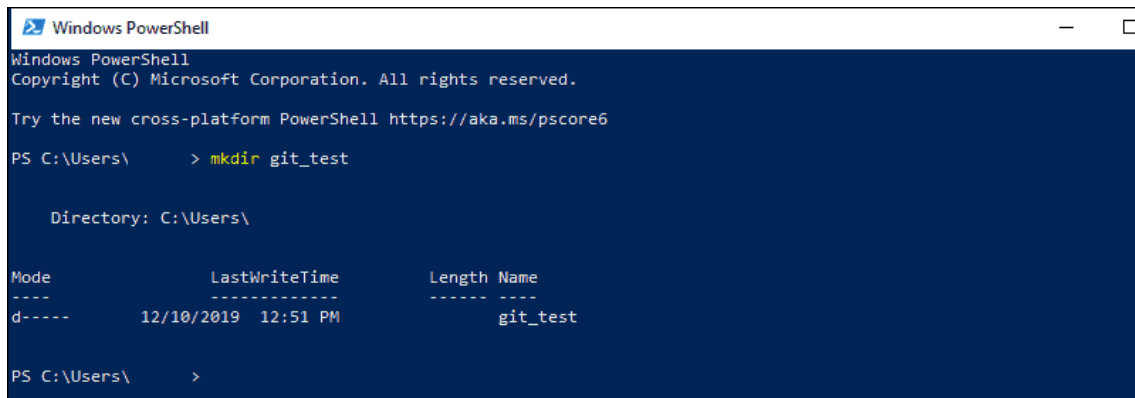
- **Create a Test Directory**

Open a Windows PowerShell interface by pressing **Windows Key + x**, and then **i** once the menu appears.

Create a new test directory (folder) by entering the following:

\$mkdir git_test

An example of the PowerShell output.



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\ > mkdir git_test

Directory: C:\Users\

Mode                LastWriteTime         Length Name
----                -
d-----         12/10/2019 12:51 PM             git_test

PS C:\Users\ >
```

Change your location to the newly created directory:

\$cd git_test

Note: If you already have a GitHub repository, use the name of that project instead of **git_test**.

3. **Configure GitHub Credentials**

Configure your local Git installation to use your GitHub credentials by entering the following:

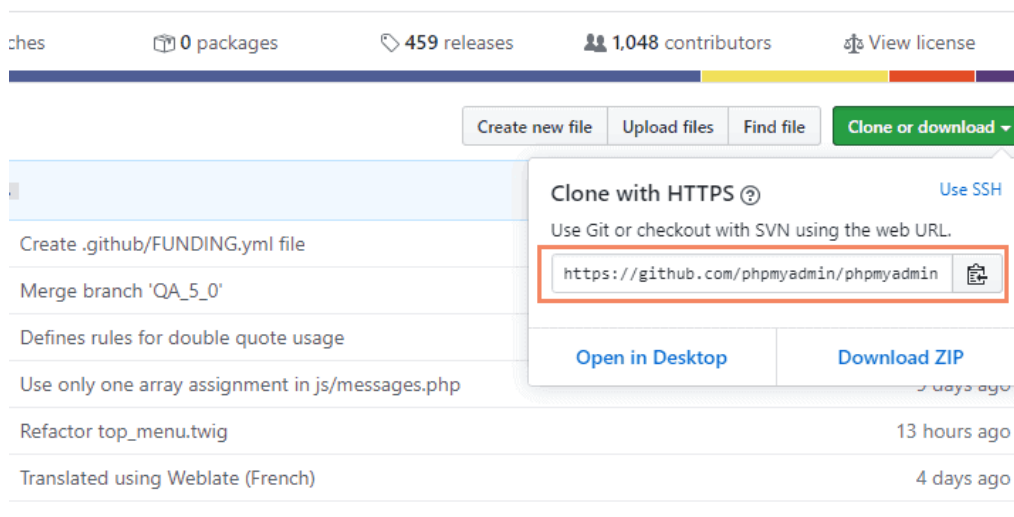
\$git config --global user.name "github_username"

\$git config --global user.email "email_address"

Note: Replace **github_username** and **email_address** with your GitHub credentials.

4. **Clone a GitHub Repository**

Go to your repository on GitHub. In the top right above the list of files, open the **Clone or download** drop-down menu. Copy the **URL for cloning over HTTPS**.



Switch to your PowerShell window, and enter the following:

\$git clone repository_url

5. List Remote Repositories

Your working directory should now have a copy of the repository from GitHub. It should contain a directory with the name of the project. Change to the directory:

\$cd git_project

Note: Replace **git_project** with the actual name of the repository you downloaded. If it's not working, you can list the contents of the current directory with the **ls command**. This is helpful if you don't know the exact name or need to check your spelling.

Once you're in the sub-directory, list the remote repositories:

\$git remote -v

1. Pushing Local Files to the Remote Repository

Once you've done some work on the project, you may want to submit those changes to the remote project on GitHub.

- a. For example, create a new text file by entering the following into your PowerShell window:

\$new-item text.txt

- b. Confirmation that the new file is created.

```
PS C:\Users\CCBill\git_test> new-item text.txt

Directory: C:\Users\CCBill\git_test

Mode                LastWriteTime         Length Name
----                -
-a-----         12/10/2019 12:58 PM              0 text.txt

PS C:\Users\CCBill\git_test>
```

- c. Now check the status of your [new Git branch](#) and untracked files:

\$git status

- d. Add your new file to the local project:

\$git add text.txt

- e. Run **git status** again to make sure the text.txt file has been added. Next, commit the changes to the local project:

\$git commit -m "Sample 1"

- f. Finally, push the changes to the remote GitHub repository:

\$git push

- g. You may need to enter your username and password for GitHub.

Expt. 06) Demonstrate RSA algorithm using Python

```
import math,sys,sympy

p = int(input("Enter the value of p:"))
q = int(input("Enter the value of q(q!=p):"))

n = p * q

phi = (p-1)*(q-1)

e = input("Enter the value of e(optional):")

if e == "":
    e = 2
    while (e<phi):
        if (math.gcd(e,phi)==1):
            break
        e += 1
elif e.isdigit():
    e = int(e)
    if(e<phi and math.gcd(e,phi) == 1):
        print("")
    else:
        sys.exit("Invalid e value")
else:
    sys.exit("Invalid Input!!!!")

d = pow(e, -1, phi)

if sympy.isprime(p)==False:
    sys.exit("Invalid p value")

elif sympy.isprime(q)==False:
    sys.exit("Invalid q value")

elif p==q:
    sys.exit("p and q should be different")

print("e = ",e, "d = ",d)

msg = input("Enter the message:")

cip = []
```

```
for char in msg:
    c = ord(char)
    c1 = pow(c,e,n)
    cip.append(c1)
print("Original Message",msg)
print("Cipher text:",cip)
plain = ""
for char in cip:
    dec = pow(char,d,n)
    dec1 = chr(dec)
    plain += dec1
print("Decrypted Message:", plain)
```

Expt. 07) Demonstrate Diffie-Hellman key exchange algorithm using Python

```
p = 23
g = 5
alice = 4
bob = 3
A = pow(g,alice,p)
B = pow(g,bob,p)
print("Alice's public key:",A)
print("Bob's public key:",B)

a_shared = pow(B,alice,p)
b_shared = pow(A,bob,p)

print(f"Alice's shared secret: {a_shared}")
print(f"Bob's shared secret: {b_shared}")
```

Expt. 08) Demonstrate Caesar Cipher using Python

```
def encode(msg, shift):
```

```
    cipher = ""
```

```
    for i in range(len(msg)):
```

```
        v = ord(msg[i])+shift
```

```
        cipher += chr(v)
```

```
    return cipher
```

```
def decode(cipher, shift):
```

```
    text = ""
```

```
    for j in range(len(cipher)):
```

```
        w = ord(cipher[j])-shift
```

```
        text += chr(w)
```

```
    return text
```

```
msg = input("Enter the message:")
```

```
shift = int(input("Enter the shift key:"))
```

```
cipher = encode(msg,shift)
```

```
print("cipher text: ", cipher)
```

```
text = decode(cipher,shift)
```

```
print("plain text: ", text)
```

Expt. 09) Study and execute different types of DOS commands.

1. Know the IP Address of any website

The ping command sends an Internet Control Message Protocol (ICMP) Request message to a remote computer to verify LAN connectivity.

C:\WINDOWS\system32>ping www.allabouttesting.com

```
C:\WINDOWS\system32>ping www.allabouttesting.org

Pinging allabouttesting.org [148.66.136.66] with 32 bytes of data:
Reply from 148.66.136.66: bytes=32 time=240ms TTL=52
Reply from 148.66.136.66: bytes=32 time=241ms TTL=52
```

2. Resolving DNS into IP

C:\WINDOWS\system32>nslookup www.allabouttesting.com

The nslookup command queries your configured DNS server to discover the IP address.

```
C:\WINDOWS\system32>nslookup www.allabouttesting.org
DNS request timed out.
    timeout was 2 seconds.
Server: UnKnown
Address: fe80::1

Non-authoritative answer:
Name:    www.allabouttesting.org
Address: 148.66.136.66
```

3. Displaying the route path between your computer to website

C:\WINDOWS\system32>tracert www.allabouttesting.com

```
C:\WINDOWS\system32>tracert www.allabouttesting.org

Tracing route to allabouttesting.org [148.66.136.66]
over a maximum of 30 hops:

  0  0 ms  0 ms  0 ms  dlink.router [192.168.0.1]
  1  11 ms  <1 ms  1 ms  abts-kk-dynamic-001.64.172.122.airtelbroadband.in [122.172.64.1]
  2   4 ms   4 ms  4 ms  abts-kk-static-069.33.166.122.airtelbroadband.in [122.166.33.69]
  3   5 ms   3 ms  2 ms  aes-static-025.102.22.125.airtel.in [125.22.102.25]
  4  11 ms   4 ms  2 ms
  5 117 ms 115 ms 132 ms 182.79.222.81
  6 206 ms 202 ms 117 ms 213.242.116.165
  7 306 ms 305 ms 305 ms 4.69.153.133
  8 282 ms 282 ms 281 ms 4.53.230.102
  9 241 ms 252 ms 240 ms ip-97-74-252-185.ip.secureserver.net [97.74.252.185]
```

4. Show the ARP table

The arp command is used to display or change entries in the ARP cache.

C:\WINDOWS\system32>arp -a

```
C:\WINDOWS\system32>arp -a

Interface: 192.168.65.1 --- 0xe
Internet Address      Physical Address      Type
192.168.65.254        00-50-56-fe-ba-18    dynamic
192.168.65.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.45.1 --- 0xf
Internet Address      Physical Address      Type
192.168.45.255        ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static

Interface: 192.168.0.102 --- 0x13
Internet Address      Physical Address      Type
192.168.0.1           10-be-f5-7c-75-f2    dynamic
192.168.0.255         ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static
```

5. Routing table, gateway, interface and metric

The route command is used to manipulate network routing tables.

C:\WINDOWS\system32>route print

```
C:\WINDOWS\system32>route print
=====
Interface List
20...70 5a 0f 08 56 39 .....Realtek PCIe FE Family Controller
17...32 f7 72 51 ba 3d .....Microsoft Wi-Fi Direct Virtual Adapter #3
15...00 50 56 c0 00 00 .....VMware Virtual Ethernet Adapter for VMnet0
14...00 50 56 c0 00 08 .....VMware Virtual Ethernet Adapter for VMnet8
19...30 f7 72 51 ba 3d .....Realtek RTL8723BE 802.11 bgn Wi-Fi Adapter
2...30 f7 72 51 ba 3e .....Bluetooth Device (Personal Area Network)
1.....Software Loopback Interface 1
11...00 00 00 00 00 00 e0 Microsoft Teredo Tunneling Adapter
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface        Metric
0.0.0.0                    0.0.0.0          192.168.0.1      192.168.0.102    55
127.0.0.0                  255.0.0.0        On-link          127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link          127.0.0.1        331
127.255.255.255            255.255.255.255  On-link          127.0.0.1        331
192.168.0.0                255.255.255.0    On-link          192.168.0.102    311
192.168.0.102              255.255.255.255  On-link          192.168.0.102    311
192.168.0.255              255.255.255.255  On-link          192.168.0.102    311
192.168.45.0               255.255.255.0    On-link          192.168.45.1     291
192.168.45.1               255.255.255.255  On-link          192.168.45.1     291
192.168.45.255             255.255.255.255  On-link          192.168.45.1     291
192.168.65.0               255.255.255.0    On-link          192.168.65.1     291
192.168.65.1               255.255.255.255  On-link          192.168.65.1     291
192.168.65.255             255.255.255.255  On-link          192.168.65.1     291
224.0.0.0                  240.0.0.0        On-link          127.0.0.1        331
```

6. See IP, gateway, dns and other info

The ipconfig command is used to display detailed IP information for each network adapter utilizing TCP/IP.

C:\WINDOWS\system32>ipconfig /all

```
C:\WINDOWS\system32>ipconfig/all

Windows IP Configuration

Host Name . . . . . : DESKTOP-90QIIIR
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : dlink.router

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe FE Family Controller
Physical Address. . . . . : 70-5A-0F-08-56-39
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 14:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
Physical Address. . . . . : 32-F7-72-51-BA-3D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
```

7. See connection status

The netstat command is used to display all open network connections and listening ports.

C:\WINDOWS\system32>netstat -an

```
C:\WINDOWS\system32>netstat -an

Active Connections

Proto Local Address           Foreign Address         State
TCP 0.0.0.0:135              0.0.0.0:0               LISTENING
TCP 0.0.0.0:445              0.0.0.0:0               LISTENING
TCP 0.0.0.0:902              0.0.0.0:0               LISTENING
TCP 0.0.0.0:912              0.0.0.0:0               LISTENING
TCP 0.0.0.0:49664            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49665            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49666            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49667            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49668            0.0.0.0:0               LISTENING
TCP 0.0.0.0:49670            0.0.0.0:0               LISTENING
TCP 192.168.0.102:139        0.0.0.0:0               LISTENING
TCP 192.168.0.102:50284      210.176.156.41:443      ESTABLISHED
TCP 192.168.0.102:50285      210.176.156.33:443      ESTABLISHED
TCP 192.168.0.102:50359      210.176.156.81:443      ESTABLISHED
TCP 192.168.0.102:50360      210.176.156.25:443      ESTABLISHED
TCP 192.168.0.102:50361      210.176.156.25:443      ESTABLISHED
TCP 192.168.0.102:50362      210.176.156.25:443      ESTABLISHED
TCP 192.168.0.102:50394      8.43.72.97:443          ESTABLISHED
TCP 192.168.0.102:50396      149.174.66.131:443      ESTABLISHED
TCP 192.168.0.102:50400      8.43.72.97:443          ESTABLISHED
TCP 192.168.0.102:50426      64.38.119.27:443        ESTABLISHED
TCP 192.168.0.102:50452      168.63.100.61:443        TIME_WAIT
TCP 192.168.0.102:65192      111.221.29.99:443        ESTABLISHED
TCP 192.168.45.1:139        0.0.0.0:0               LISTENING
TCP 192.168.65.1:139        0.0.0.0:0               LISTENING
TCP [::]:135                [::]:0                  LISTENING
TCP [::]:445                [::]:0                  LISTENING
TCP [::]:49664              [::]:0                  LISTENING
```

8. Display or set a specific path available to executable files

C:\WINDOWS\system32>path

```
C:\WINDOWS\system32>path
PATH=C:\ProgramData\Oracle\Java\javapath;C:\WINDOWS\system32;C:\WINDOWS;C:\WINDOWS\System32\Wbem;C:\WINDOWS\System32\WindowsPowerShell\v1.0\;C:\Users\Geeta\AppData\Local\Microsoft\WindowsApps
```

9. System File Checker is an automatic scan and repair tool mainly for Windows system files

C:\WINDOWS\system32>**sfc /scannow**

```
Microsoft Windows [Version 10.0.15063]
(c) 2017 Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>sfc /scannow

Beginning system scan. This process will take some time.

Beginning verification phase of system scan.
Verification 5% complete.
```

10. Wi-Fi password

If you forgot your wi-fi password, use below command replacing SSID with the name of the network

C:\WINDOWS\system32> **netsh wlan show profile SSID key=clear**

If you're not sure about the networks you've connected to, use below command to find out wi-fi network

C:\WINDOWS\system32> **netsh wlan show profile**

```
C:\WINDOWS\system32>netsh wlan show profile Clarksinn key=clear

Profile Clarksinn on interface Wi-Fi:
=====

Applied: All User Profile

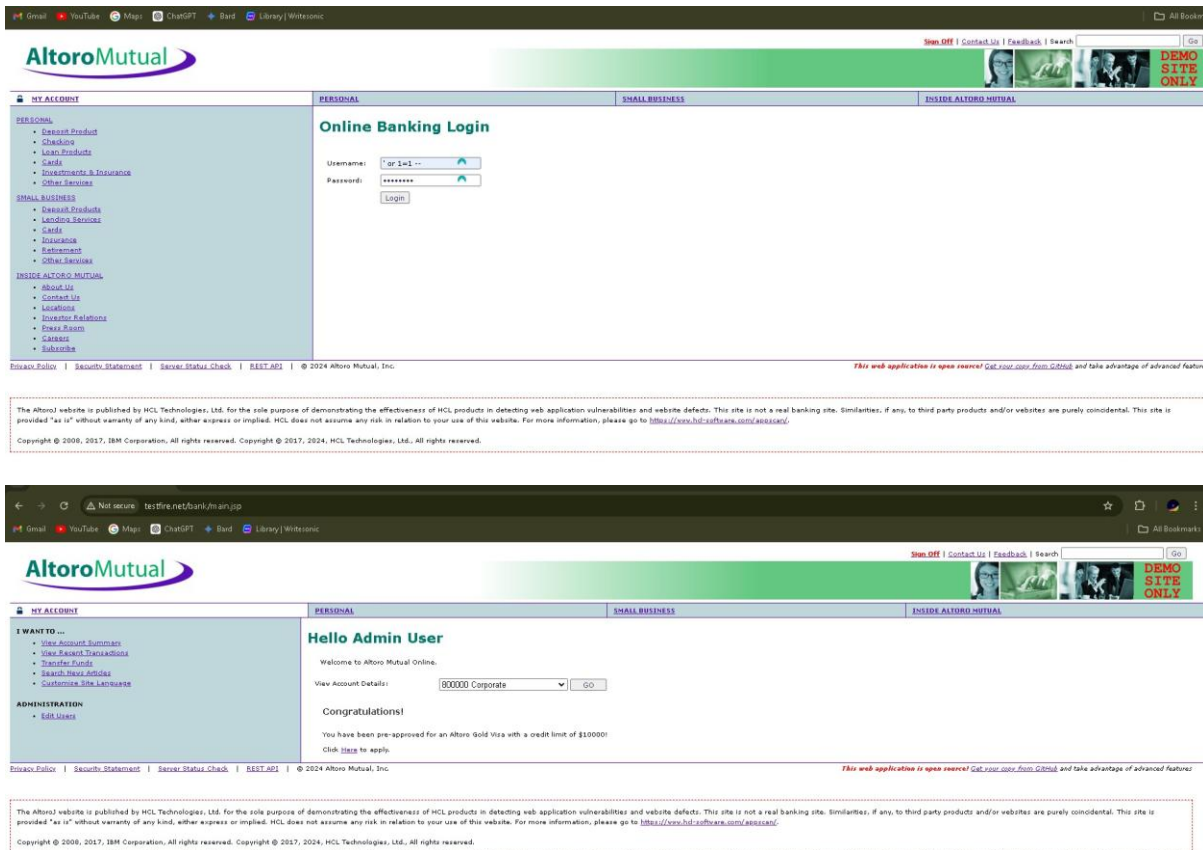
Profile information
-----
Version                : 1
Type                   : Wireless LAN
Name                   : Clarksinn
Control options        :
  Connection mode       : Connect manually
  Network broadcast     : Connect only if this network is broadcasting
  AutoSwitch            : Do not switch to other networks
  MAC Randomization     : Disabled

Connectivity settings
-----
Number of SSIDs        : 1
SSID name              : "Clarksinn"
Network type           : Infrastructure
Radio type             : [ Any Radio Type ]
Vendor extension        : Not present

Security settings
-----
Authentication         : WPA2-Personal
Cipher                 : TKIP
Security key            : Present
Key Content             : welcome@clarksinn

Cost settings
-----
Cost                   : Unrestricted
Congested              : No
Approaching Data Limit : No
Over Data Limit        : No
Roaming                : No
Cost Source            : Default
```

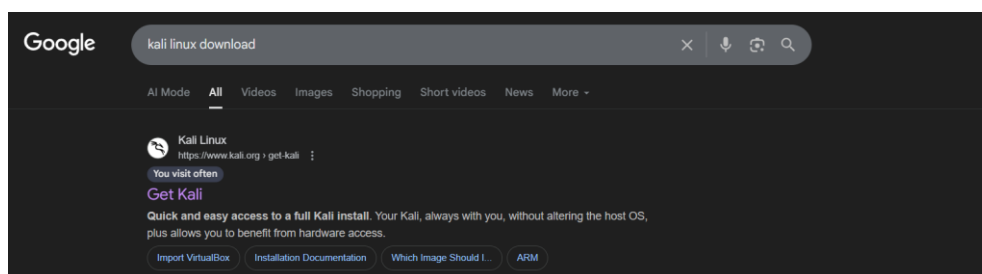

STEP 4: Then we write an SQL script in the username field and we write the random text in the password field. We use the SQL script ' or 1=1--. In this script, the 1=1 condition is always true. The -- symbol comments out the rest of the query, meaning the password field will be ignored.



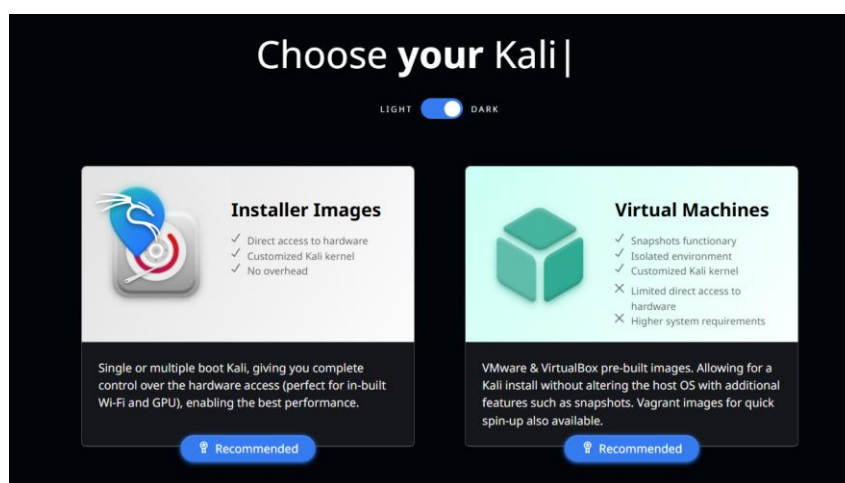
Expt. 11) Demonstrate SELinux (Hardening of Linux)

Kali Linux installation steps:

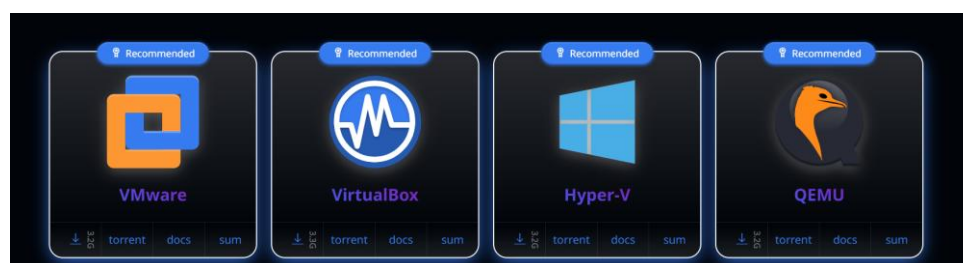
1. Open Google Chrome: Launch your Google Chrome browser.
2. Search for Kali Linux Download: In the address bar, type "Kali Linux Download" and press Enter.
3. Open the First Website: Click on the first search result, which should be the official Kali Linux website.



4. Choose Your Platform: On the Kali Linux website, look for a section or tab labeled "Downloads" or "Get Kali" and click on it.
5. Click on Virtual Images: Find the option for "Virtual Images" or similar, and click on it. This section is where you'll find images designed for use with virtual machines.



6. Download the Virtual box: Locate the download button or link for the virtual image (usually an icon) and click it to start downloading the image file.



SELinux stands for Security-Enhanced Linux, which is an access control system that is built into the Linux kernel. It is used to enforce their security policies that define what level of access users, programs, and services have on a system.

SELinux Operating Modes

SELinux can operate in two global modes:

- Permissive mode, in which permission denials are logged but not enforced.
- Enforcing mode, in which permissions denials are both logged and enforced.

Setting SELinux Modes SELinux runs in one of three modes

- 1) Disabled: The kernel uses only DAC rules for access control. SELinux does not enforce any security policy because no policy is loaded into the kernel.
- 2) Enforcing: The kernel denies access to users and programs unless permitted by SE Linux security policy rules. All denial messages are logged as AVC (Access Vector Cache) denials. This is the default mode that enforces SELinux security policy.
- 3) Permissive: The kernel does not enforce security policy rules but SELinux sends denial messages to a log file. This allows you to see what actions would have been denied if SELinux were running in enforcing mode. This mode is intended to be used for diagnosing the behaviour of SELinux.

Commands to Execute SELinux

- To install SELinux package in Linux system

\$ sudo apt update

\$ sudo apt install policycoreutils selinux-utils selinux-basics

- Next command to enable SELinux on system (execute this command in root terminal)

\$ sudo selinux-activate

- Next, reboot system to apply changes

\$ reboot

- Next set/change SELinux to enforcing mode

\$ sudo selinux-config-enforcing

- Next reboot system to apply changes

\$ reboot

- Next check the status of selinux

\$ sestatus

- To check mode of SE Linux

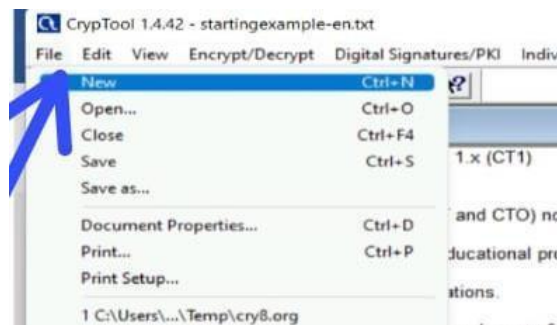
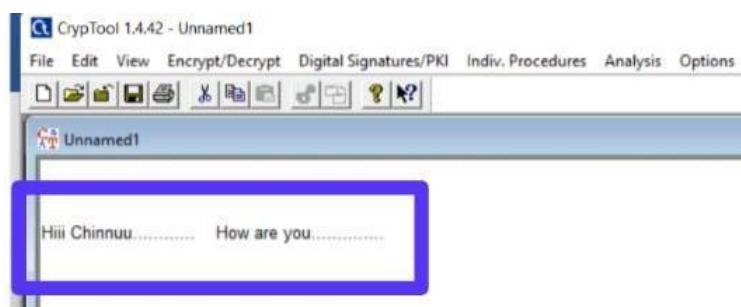
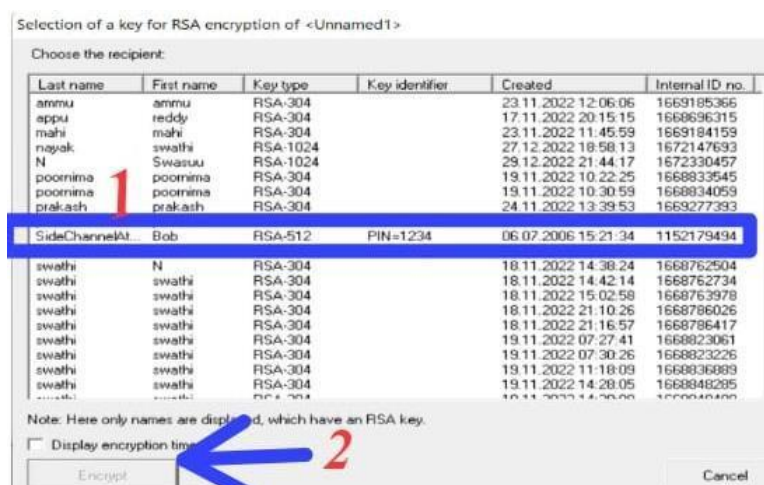
\$ getenforce

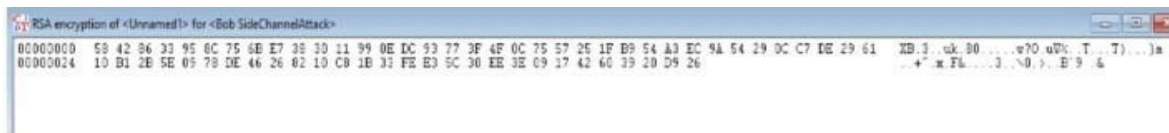
- To set the current mode to Enforcing:

\$ sudo setenforce enforcing

- To set the current mode to Permissive:

\$ sudo setenforce permissive

Expt. 12) Design a simple crypto system [encryption, decryption, digital signature] using any crypto tool**Steps for Encryption****Step 01:** Go to **File** & click on the **New** in crypto tool**Step 02:** Then type the message**Step 03:** Go to **Encryption/Decryption** option & select **Asymmetric** then click on the **RSA Encryption****Step 04:** In **Choose the recipient**, double click on the **recipient name** and then click on the **Encrypt****Step 05:** Then your message will be encrypted

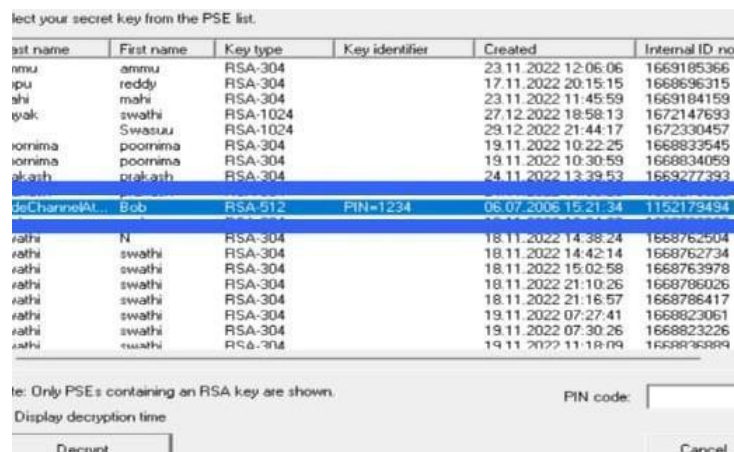


Steps for Decryption

Step 01: Go to **Encryption/Decryption** & select the **Asymmetric** then click on the **RSA decryption**



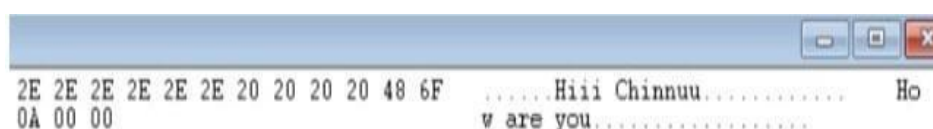
Step 03: Select your secret key from the **PSE list**, then double click on the PSE



Step 04: Enter the PIN code, then click on the **Decrypt**

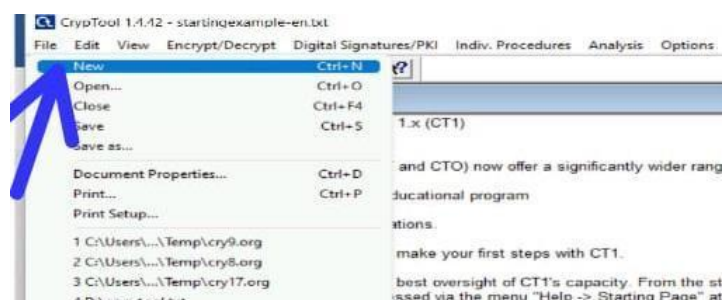


Step 05: Then your message will be decrypted

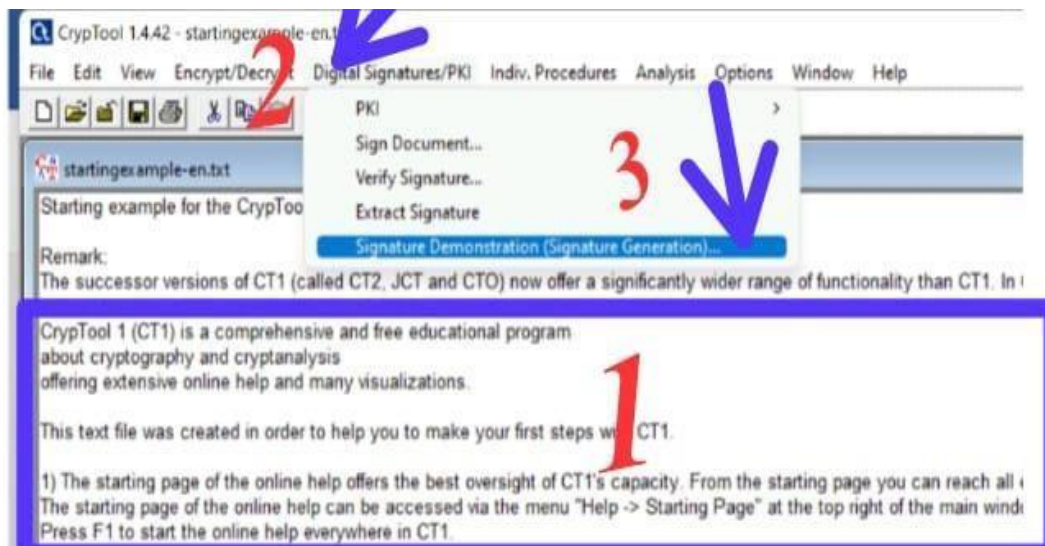


Steps for Digital Signature

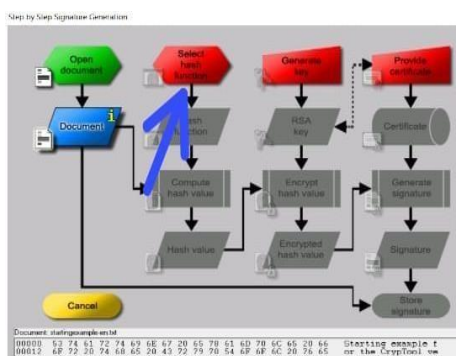
Step 01: Go to **File**, then click on the **New**



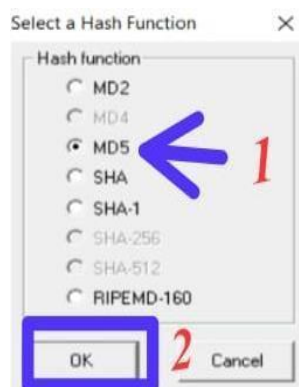
Step 02: Type the message & go to **Digital Signature** then click on the **Signature Demonstration**.



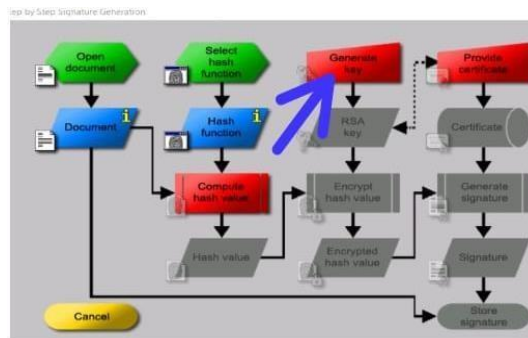
Step 03: Click on the **Select Hash Function**



Step 04: Click on **MD5** then click on **OK** option



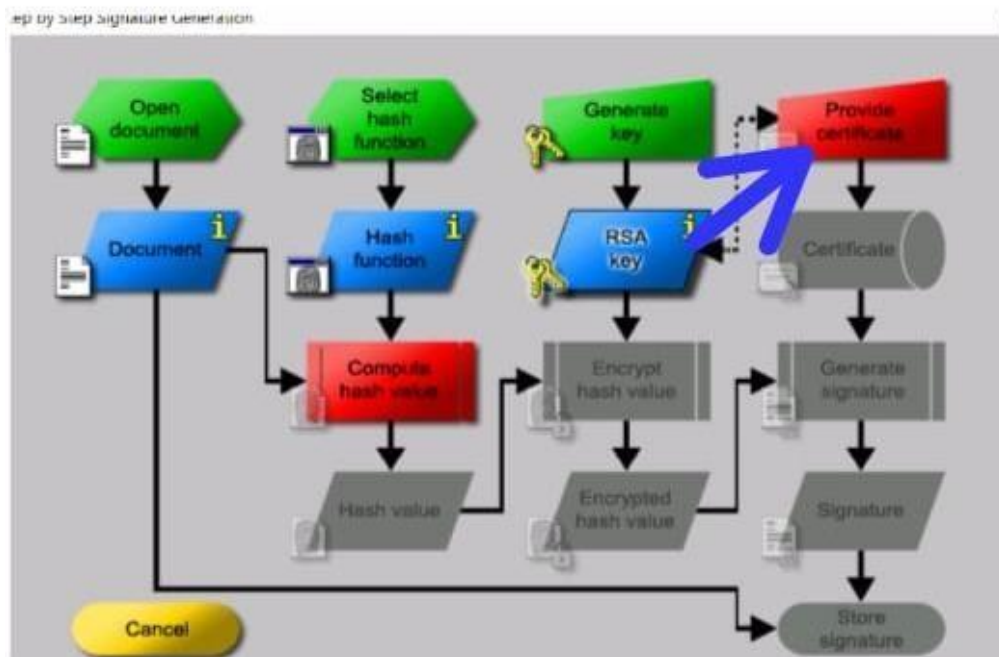
Step 05: Then click on the **Generate Key**



Step 06: Click on the **Generate prime numbers...**

Step 07: Select on the **Generate prime numbers** and then click on **Apply primes**

Step 09: Click on the **Provide Certificate**



Step 10: Create your **PSE certificate**, then click on the **Create Certificate PSE**

Create Certificate and PSE

Public RSA parameter

Bit length: 304 bit

RSA modulus N: 347554015224515657567259565554117466819209800411333107790091

Public key e: 65537

Personal data for the certificate

Name: Swasuu

First name: reddy

Key identifier: (optional)

PIN: ****

PIN verification: ****

Generated names for PSE and certificate

User Key ID: [Swasuu][reddy][RSA-304][1672332813]

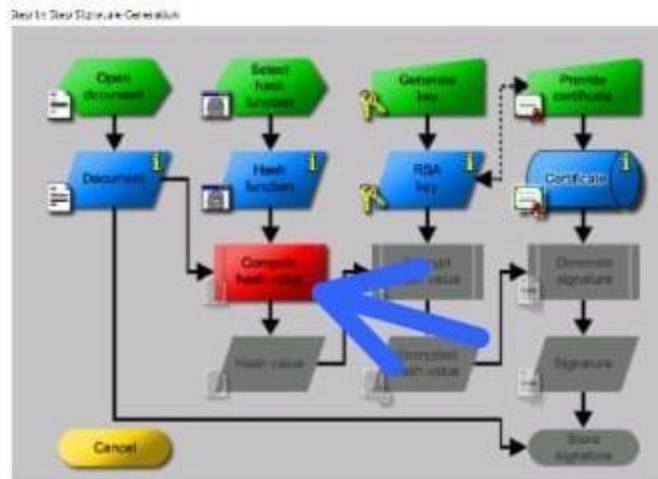
Distinguished Name: [CN=reddy Swasuu [1672332813], C=cryptool, DC=org]

Create Certificate and PSE

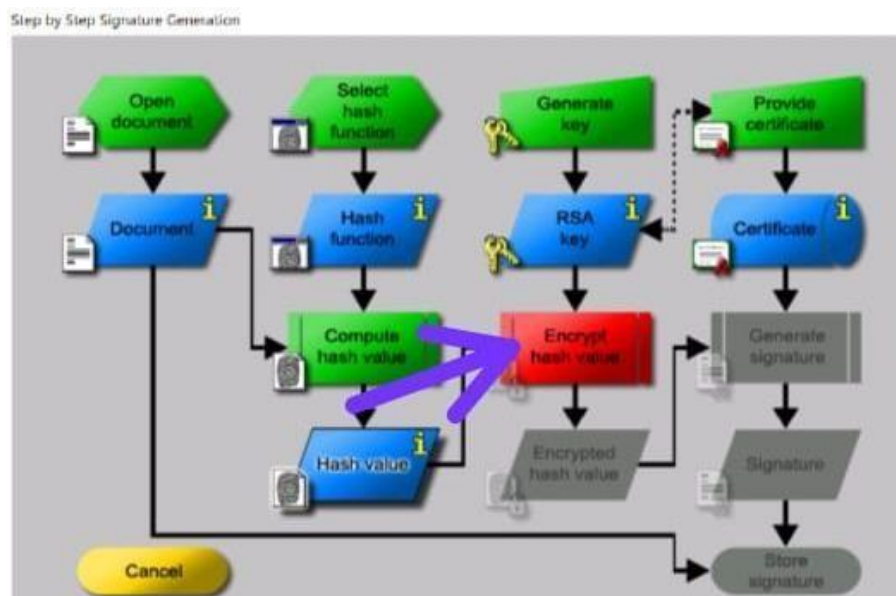
Import certificate and key

Cancel

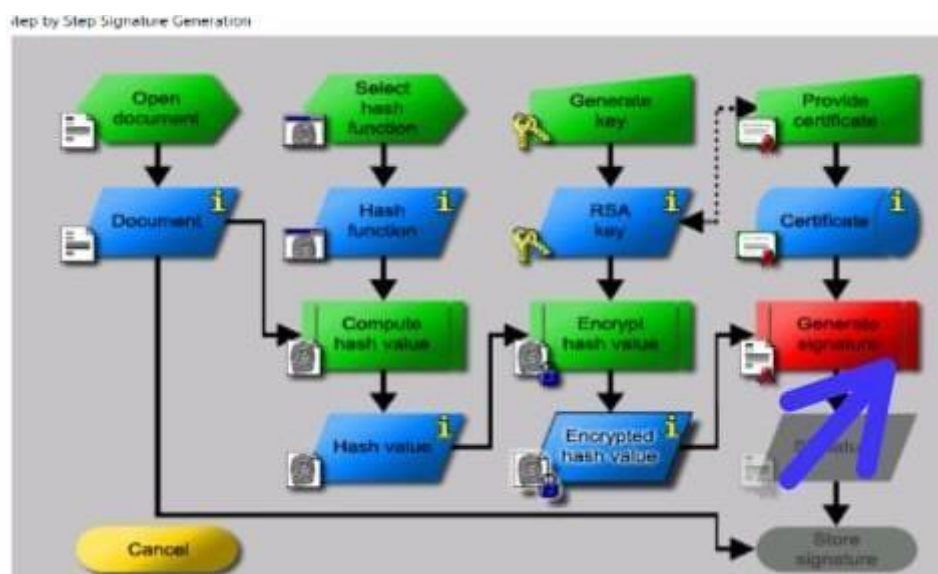
Step 11: Then click on the **Compute Hash Value**



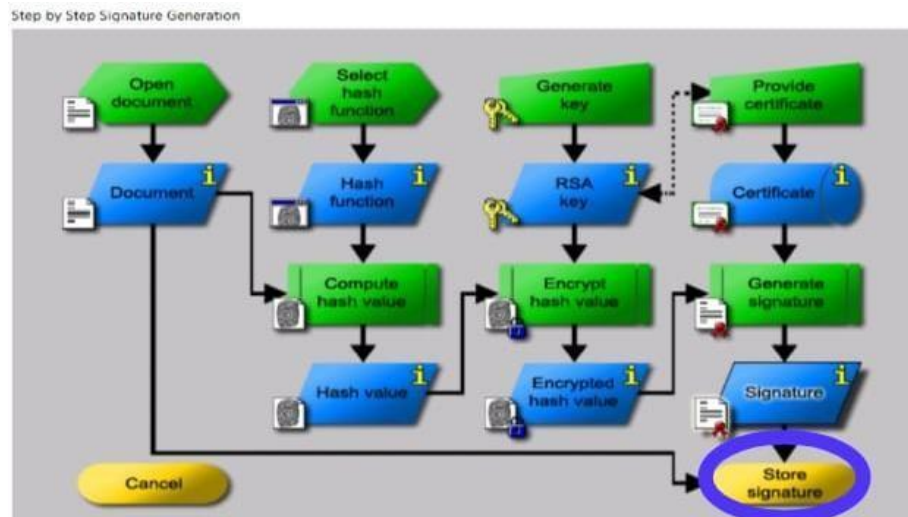
Step 12: Click on the **Encrypt Hash Value**



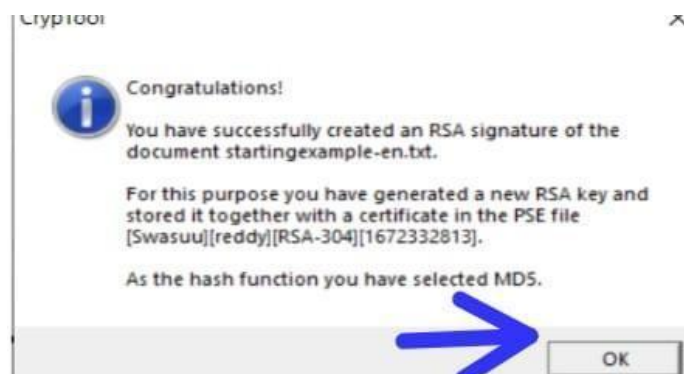
Step 13: Then click on the **Generate Signature**



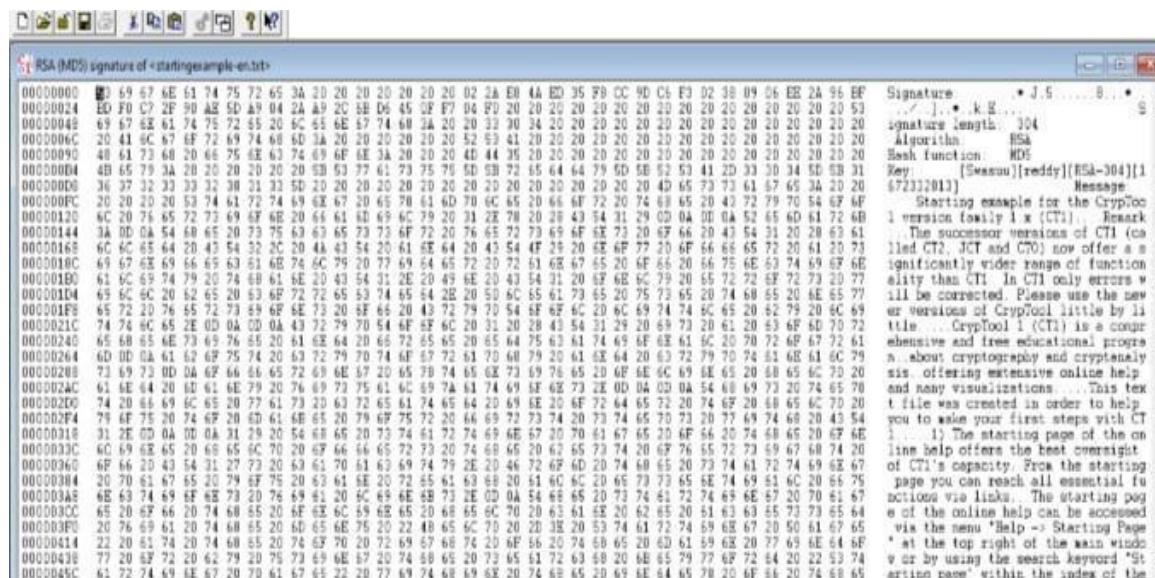
Step 14: Click on the **Store Signature**

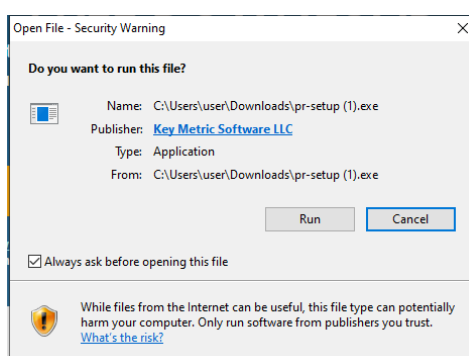


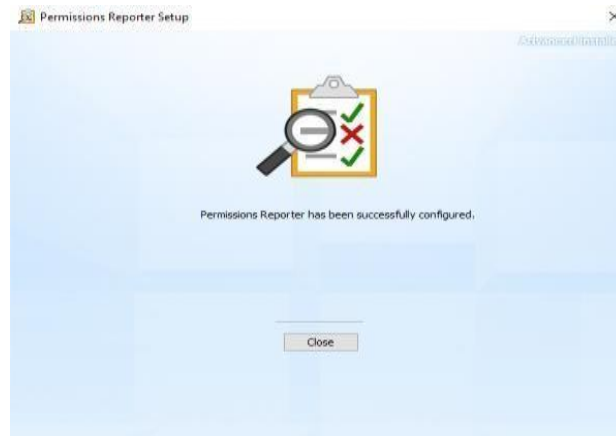
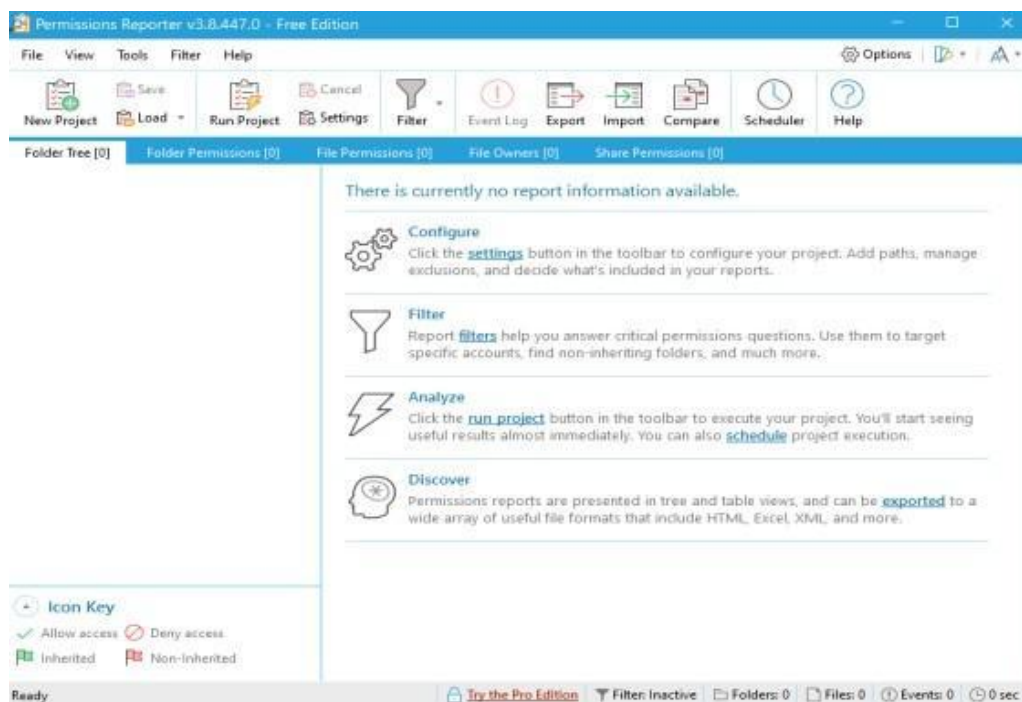
Step 15: Shows the result, then click on ok option

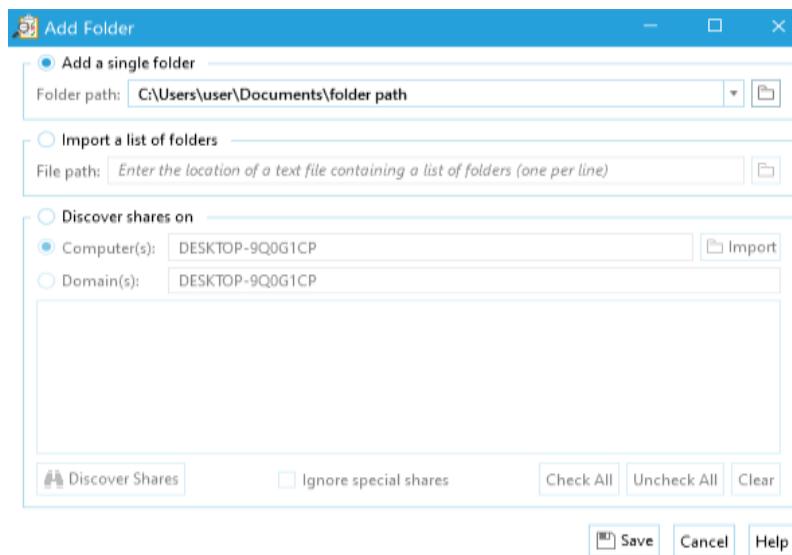


Step 16: digital signature will be displayed on the screen.

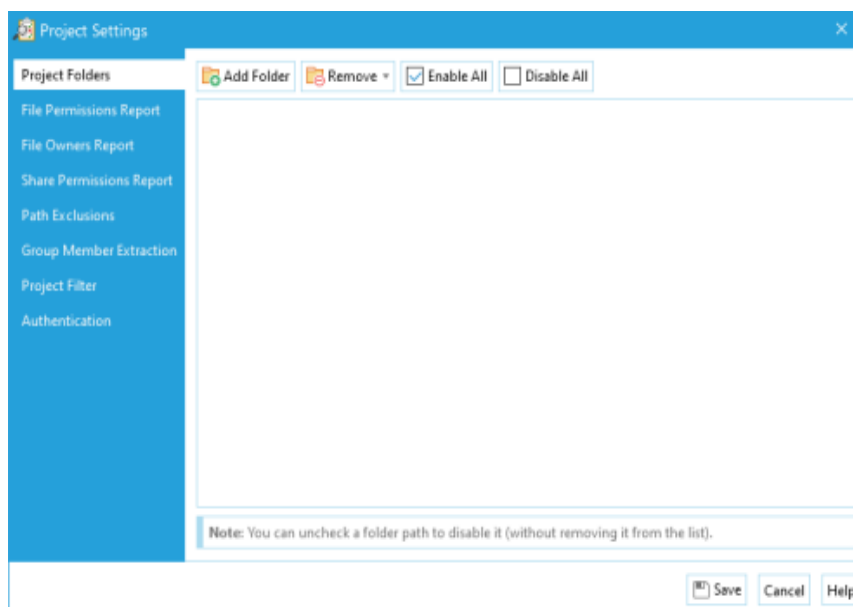


Expt. 13) Demonstrate NTFS file system using NTFS permission reporter.**Step 01:** Open any browser, search for **NTFS File Permission Reporter****Step 02:** Click on **Download Now****Step 03:** Click on **Start Download****Step 04:** Click **Run****Step 05:** Click on **Next**

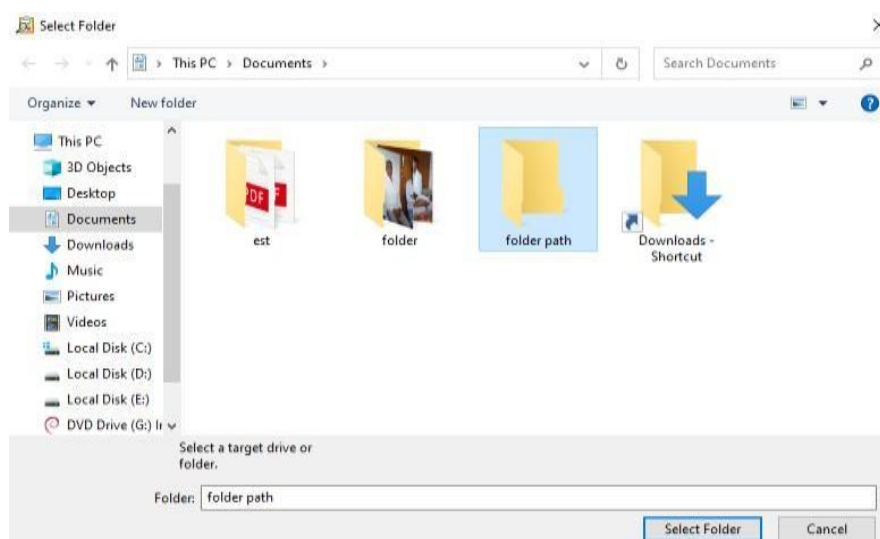
Step 06: Click on Close**Step 07: Open the Permission Reporter****Step 08: In Configure, click on Settings**

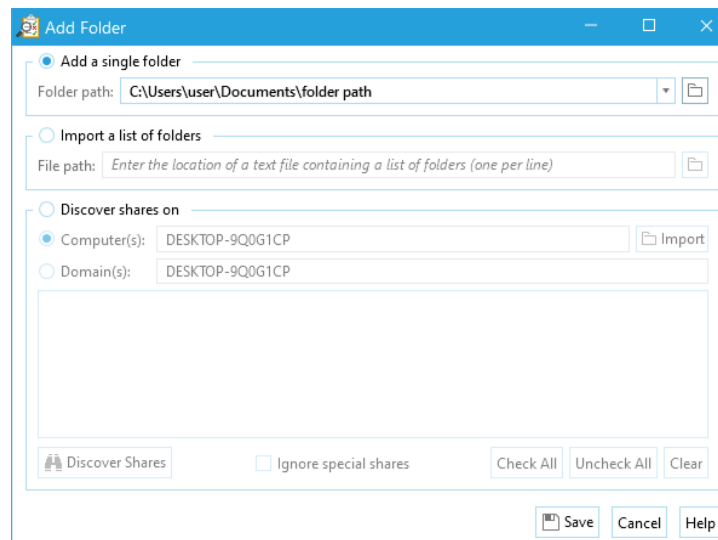
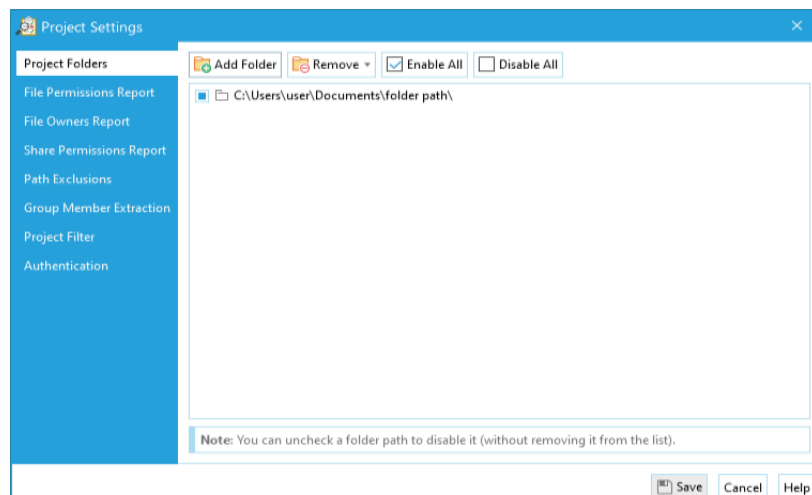
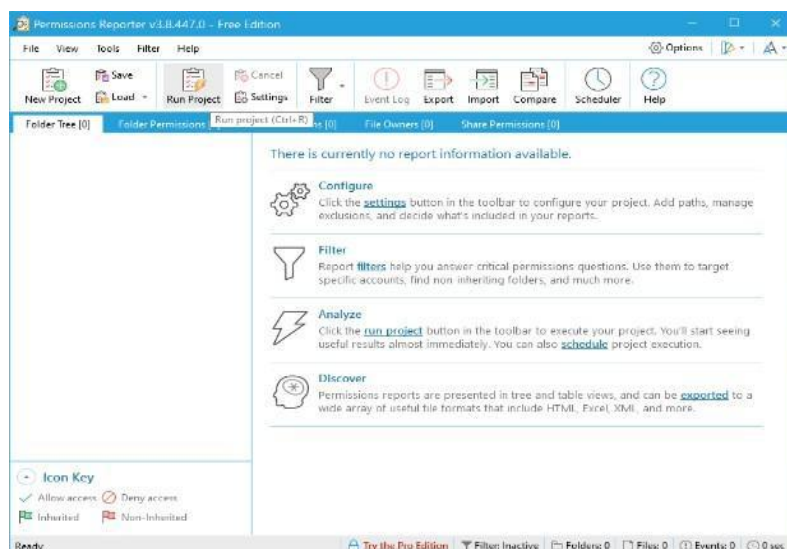


Step 09: In project settings, **Select Add Folder**

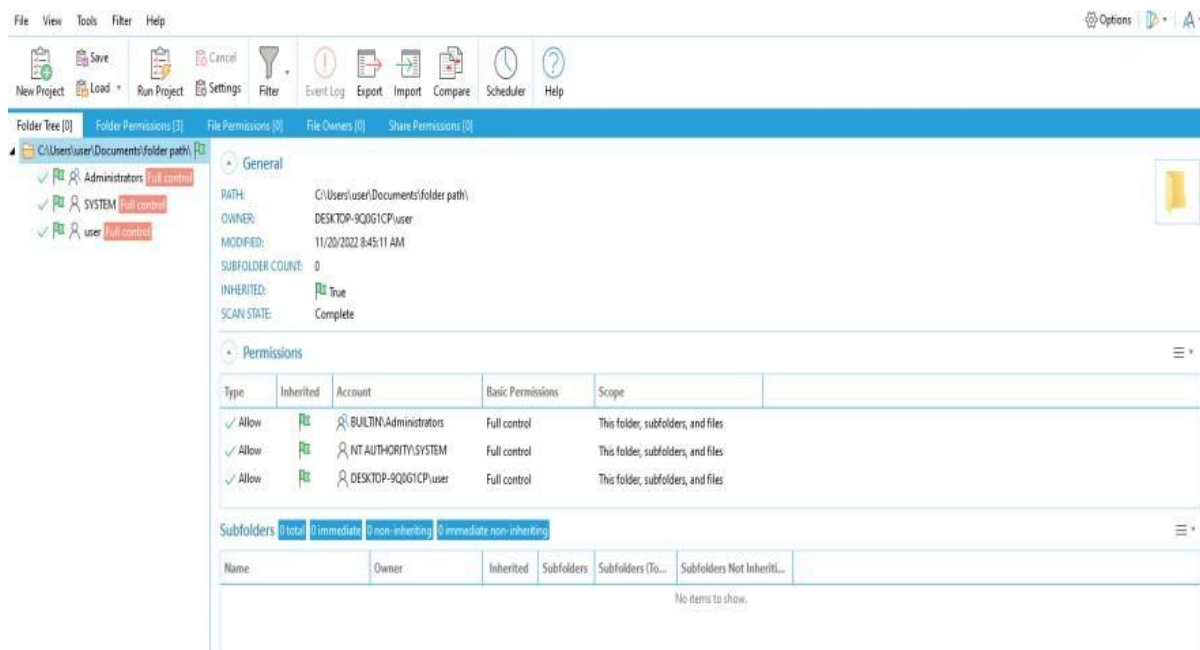


Step 10: Select the folder and click on **Select Folder**



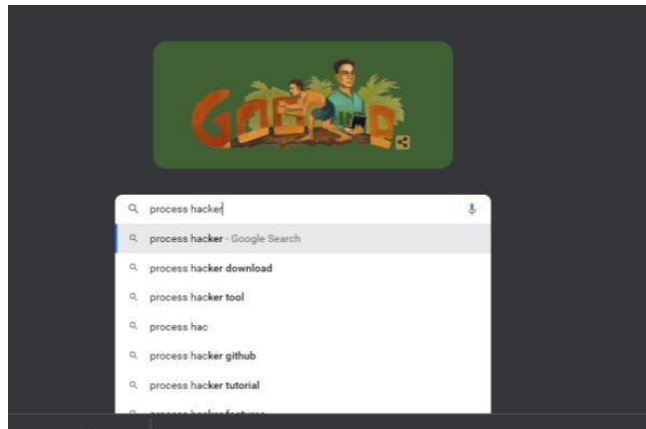
Step 11: Click on Save**Step 12: Click on Save****Step 13: Click on the Run Project**

Step 14: Detailed information of permissions of the selected folder will be displayed on the screen

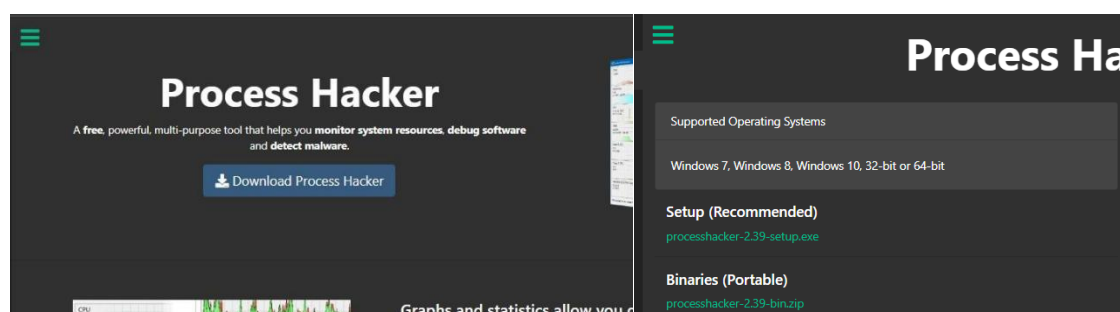


Expt. 14) Installation of process hacker and observe the process with all details.

Step 01: Go to any browser & search **Process Hacker**



Step 02: Download process hacker



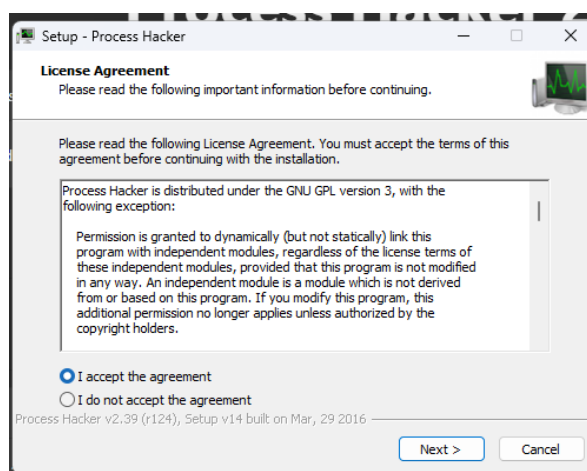
Step 03: Then click on the **processhacker-2.39-setup.exe**

Step 04: Process hacker will be downloaded

Step 05: Then process hacker will be displayed on the screen

Step 06: Double click on the process hacker setup file.

Step 07: Then click on **Yes** option and click on the **Next**



Step 08: Then click on the Install

Step 09: Click on the finish button

Expt. 15) Using the Microsoft threat model software, create a threat model for any application architecture.

Downloading and installing the software

Step 01: Go to any browser, search for “**Microsoft Threat Modeling Tool Download**”

Step 02: Select the first search result related to download from Microsoft official website.

Step 03: Click on Download button to start downloading of ThreatModelingTool2016.msi 4.0

Step 04: Goto the Downloads folder and double click the downloaded file.

Step 05: Follow the instructions and finish the setup.

Creating a new threat model

Step 01: Start **Microsoft Threat Modeling Tool**

Step 02: Goto **View** menu and select **Stencils**

Step 03: From Stencils, add **Human User**, **Web Service** and **SQL Database**

Step 04: From Stencils, drag two number of **HTTPS** connection and connect **Human User** to **Web Service**, **Web Service** to **Human User**.

Step 05: From Stencils, drag two number of **Generic Data Flow** connections and connect **Web Service** to **SQL Database** and **SQL Database** to **Web Service**.

Step 06: Select every connection and provide meaningful names in Properties.

Step 07: From Stencils, drag Internet Boundary line and use it to cover Human User to indicate that the human user’s connection is from global internet.

Step 08: From Stencils, draw the Generic Trust Border Boundary and resize it to cover Web Service and SQL database.

Step 09: Save the diagram.

Diagram of threat model:

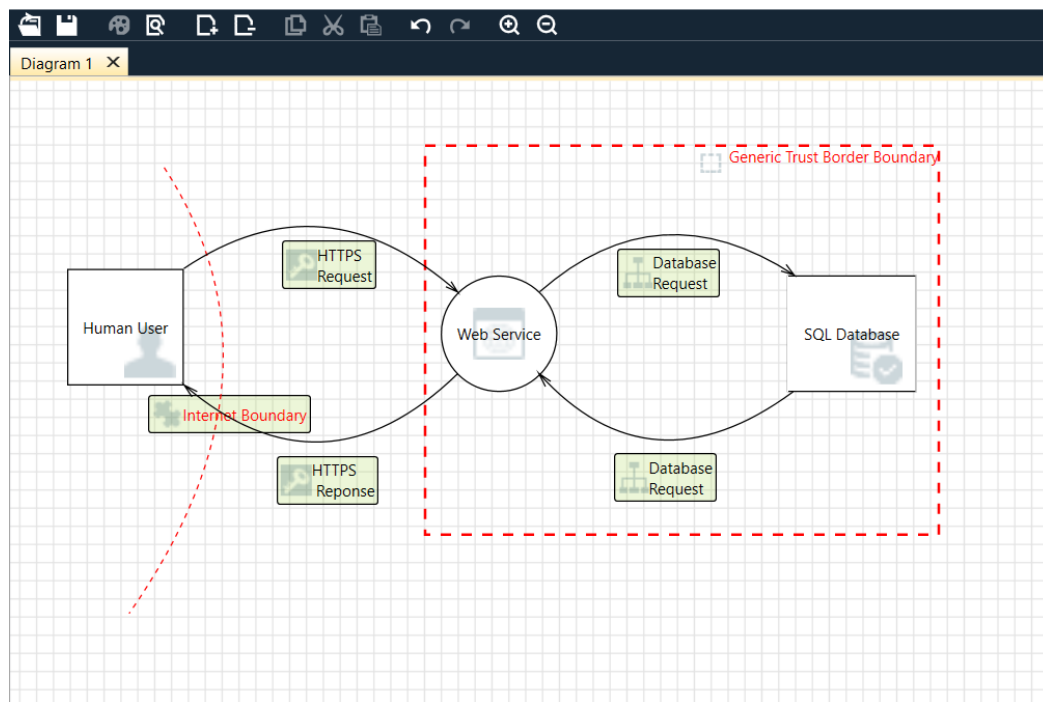


Fig: Diagram of threat model

Step 10: Using Element Properties, assign following Properties to **SQL Database**

Stores Credentials	Yes
Stores Log Data	Yes
Encrypted	Yes
Write Access	Yes
Backup	Yes

Step 11: Using Element Properties, assign following Properties to **Web Service**

Running as	System
Isolation Level	Low Integrity Level
Accepts Input From	Any Remote User or Entity
Implements or uses an authentication mechanism	Yes
Implements or uses an authorization mechanism	Yes
Implements or uses an communication protocol	Yes
Sanitizes input	Yes
Sanitizes output	No

Step 11: Using Element Properties, assign following common Properties to **Database Request** and **Database Response**

Source Authenticated	Yes
----------------------	-----

Destination Authenticated	Yes
Provides Confidentiality	Yes
Provides Integrity	Yes
Contains Cookies	Yes

Step 12: Keep the properties of HTTPS elements as those are predefined

Step 13: Goto **View** menu and select **Analysis View**. A details list of threats will be generated.

The screenshot displays the MyThreatModel - Threat Modeling Tool 2016 interface. The top section shows a threat diagram with components: Human User, Web Service, and SQL Database. Interactions include HTTPS Request, HTTPS Response, Database Request, and Database Response. Boundaries for Internet, Generic Trust, and Database are indicated. The bottom section shows a 'Threat List' table with 7 threats displayed out of 25 total.

ID	Diagram	Changed By	Last Modified	State	Title	Category	Description	Justification
3	Diagram 1	PRAMOD\pra...	18-09-2023 12:...	Not Started	Spoofing the...	Spoofing	Human User m...	
4	Diagram 1		Generated	Not Started	Elevation Usin...	Elevation Of Pr...	Web Service m...	
5	Diagram 1		Generated	Not Started	Spoofing of D...	Spoofing	SQL Database...	
6	Diagram 1		Generated	Not Started	Potential SQL I...	Tampering	SQL injection i...	
7	Diagram 1		Generated	Not Started	Potential Exces...	Denial Of Servi...	Does Web Ser...	

25 Threats Displayed, 25 Total

Threat Properties

ID: 3 Diagram: Diagram 1 Status: Not Started Last Modified: PRAMOD\pramo 18-09-2023 12:40:55

Title: Spoofing the Human User External Entity

Category: Spoofing

Description: Human User may be spoofed by an attacker and this may lead to unauthorized access to Web Service. Consider using a standard authentication mechanism to identify the external entity.

Justification:

Threat Properties Notes - no entries

Step 14: Goto Reports menu and click Create Full Report, Generate Report and save the report. The saved report will be in html format and open the report in any web browser and analyze the list of threat provided in the report.

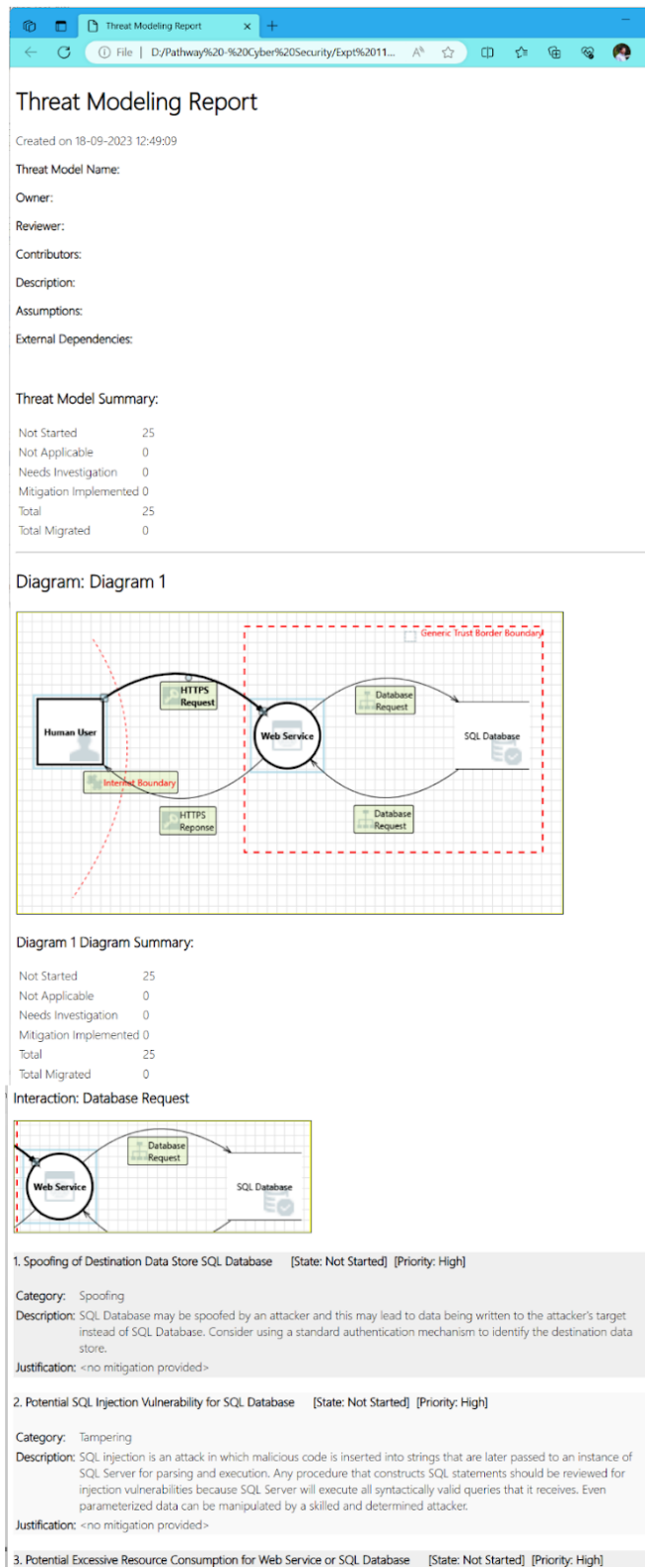


Fig: partial view of report of vulnerabilities and threat

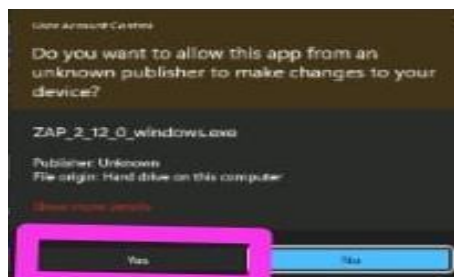
Expt. 16) Install OWASP ZAP and demonstrate finding vulnerabilities in web application using automated scan & Manual Scan.

Step 1: - Open any web browser and paste the website URL <https://www.zaproxy.org/download/>.

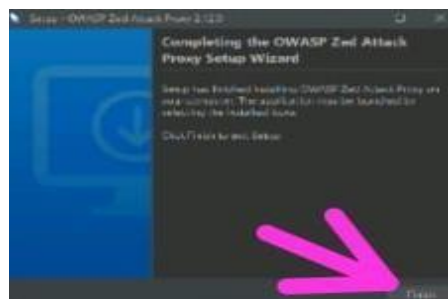
Step 2: - Select **Windows (64) Installer** then click on the Download



Step 3: - Now ZAP will be downloaded & double click on ZAP then click on yes



Step 4: - Follow the installation instructions and complete the installation process.



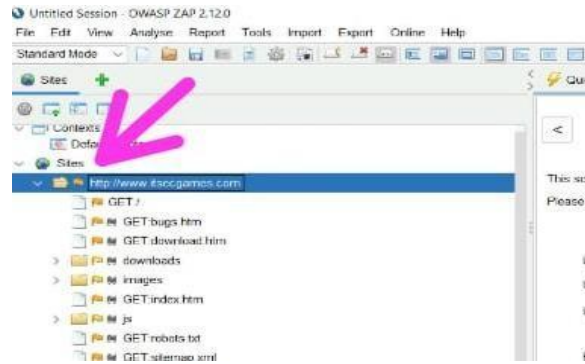
Automated scan:

Step 1: - Open the OWASP ZAP then click on the Automated scan

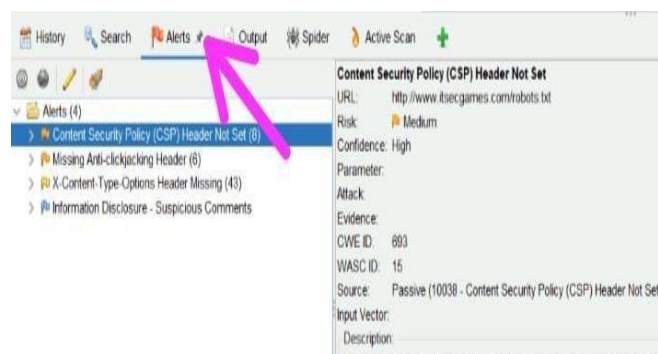
Step 2: - Paste **BW-app website's** URL <http://www.itsecgames.com/> then click on attack button



Step 3: - Click on Sites for different files in that and analyze request and response header

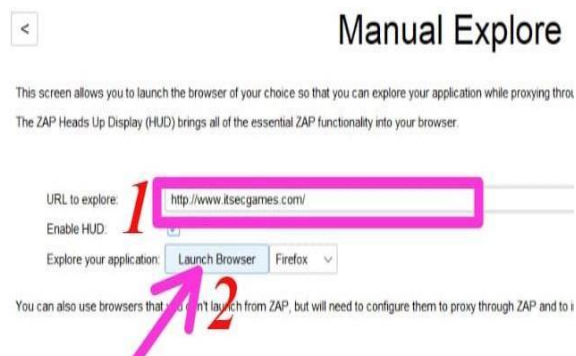


Step 4: - Now click on Alerts section and analyze type of risk which was exposed by automated scan



Manual Explore:

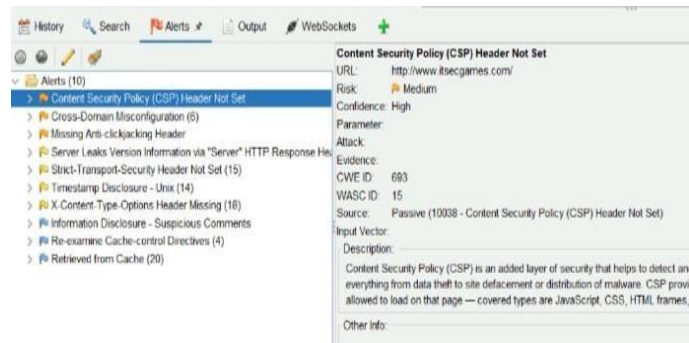
Step 1: - Open the **OWASP ZAP** & click on **Manual Explore** to manually scanning website and paste BW app URL <http://www.itsecgames.com/>. Then click on **Launch Browser**



Step 2: - Now opens BW app website



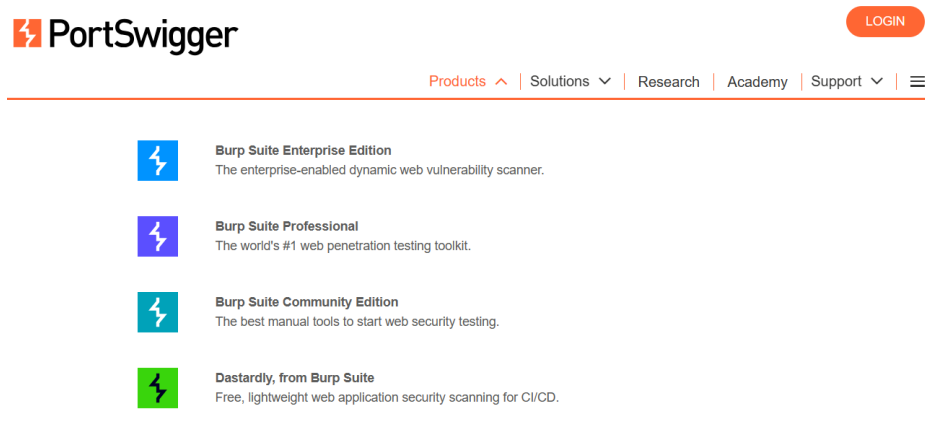
Step 3: - From website we can start spider and manual scanning. Now come Back to ZAP and check for alerts.



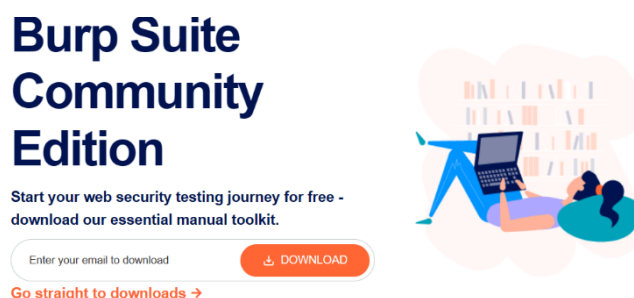
Expt. 17) Installing Burp Suite on Windows & import proxy server CA certificate to browser**Downloading and installing the Burp Suite Community Edition**

Step 01: Visit the official Burp Suite website <https://www.portswigger.net> using any web browser.

Step 02: Click on **Products**, choose **Burp suite Community Edition** as it is free, click on it.



Step 03: Click on **Go straight to downloads**.



Step 04: Select **Burp Suite Community Edition** and select **Windows (64-bit)** and then click on download.



Step 05: Now check for the downloaded file in Downloads folder and double click it to initiate the installation of software.

Step 06: Follow the instructions during the installation process and complete the process.

Step 07: Burp suite will be successfully installed on the system and an icon is created on the desktop



First run of Burp Suite Community Edition

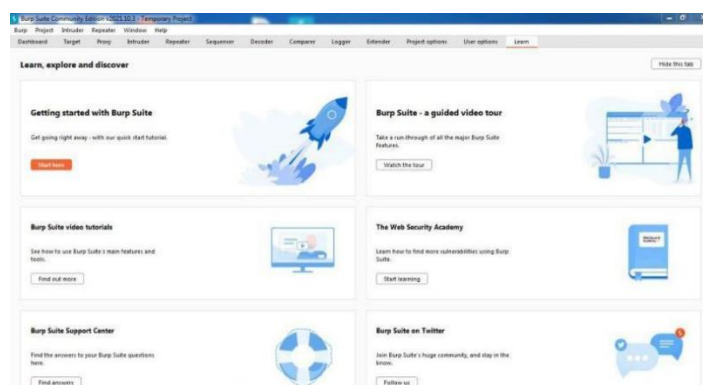
Step 01: Run the software by clicking the **Burp Suite** icon on the desktop and click on **I Accept**.

Step 02: There is only one option i.e. **Temporary project**, select it and click on **Next**.

Step 03: click on **Use Burp Defaults** and click on **Start Burp** button

Step 04: Project will start loading.

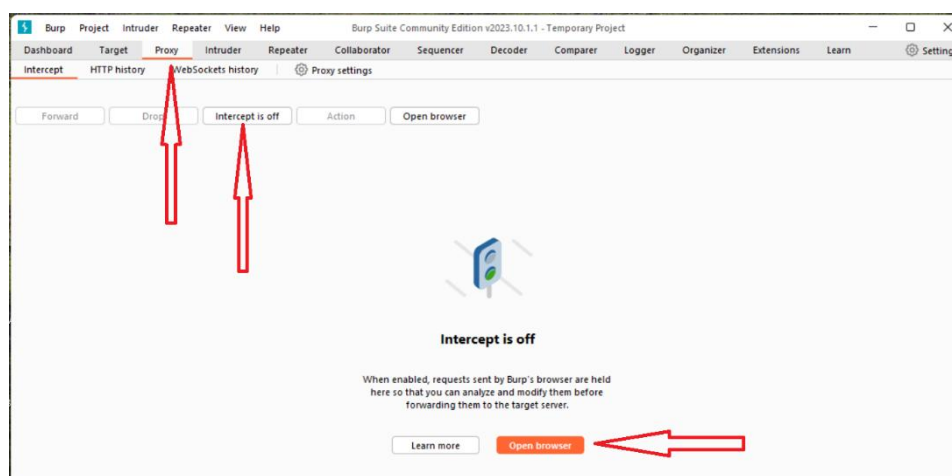
Step 05: Finally, a new project window will appear.



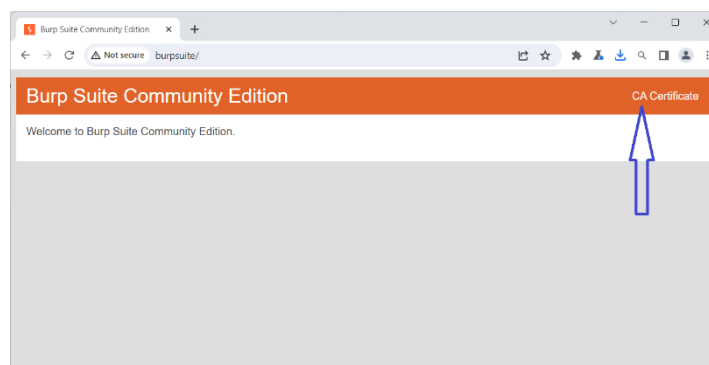
Importing Certificate to Firefox browser

Step 01: Start the **Burp Suite** application and accept **Temporary project** and **Use Burp Defaults**.

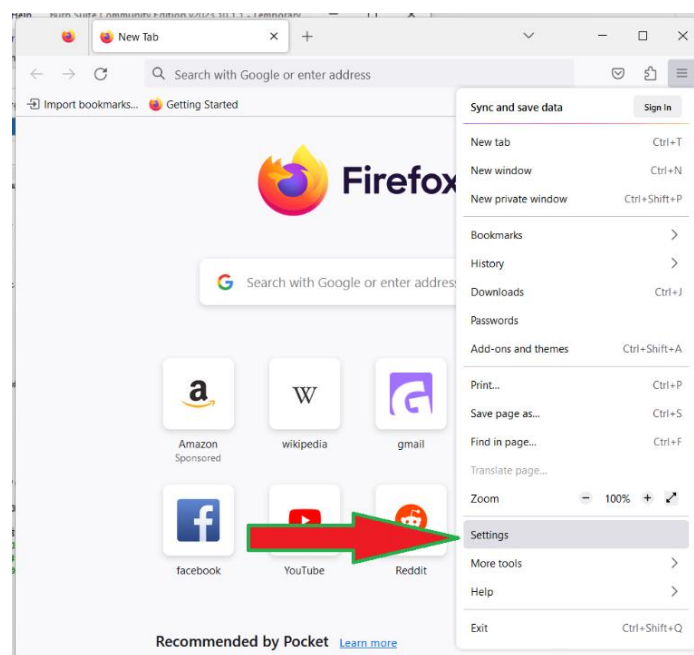
Step 02: Select the **Proxy** and turn on the **Intercept** then click on the **Open Browser** option to start a new browser window from **Burp Suite**.



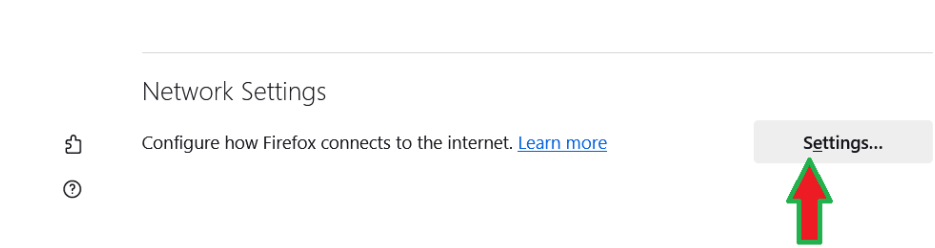
Step 03: In the browser opened by **Burp Suite**, enter the URL <http://Burpsuite>. In the opened website, click the link **CA Certificate** to download it. The default filename of certificate is **cacert.der**



Step 04: Open the **Firefox** browser, open **application menu** by clicking on **3 vertical line** and go to **Settings**.



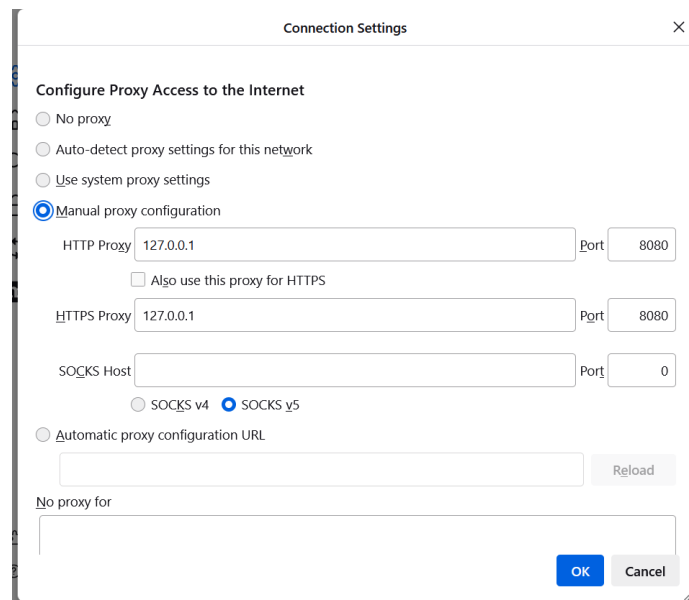
Step 05: In the **Settings** windows, scroll down to last and in the **Network Settings** section, click on **Settings** button



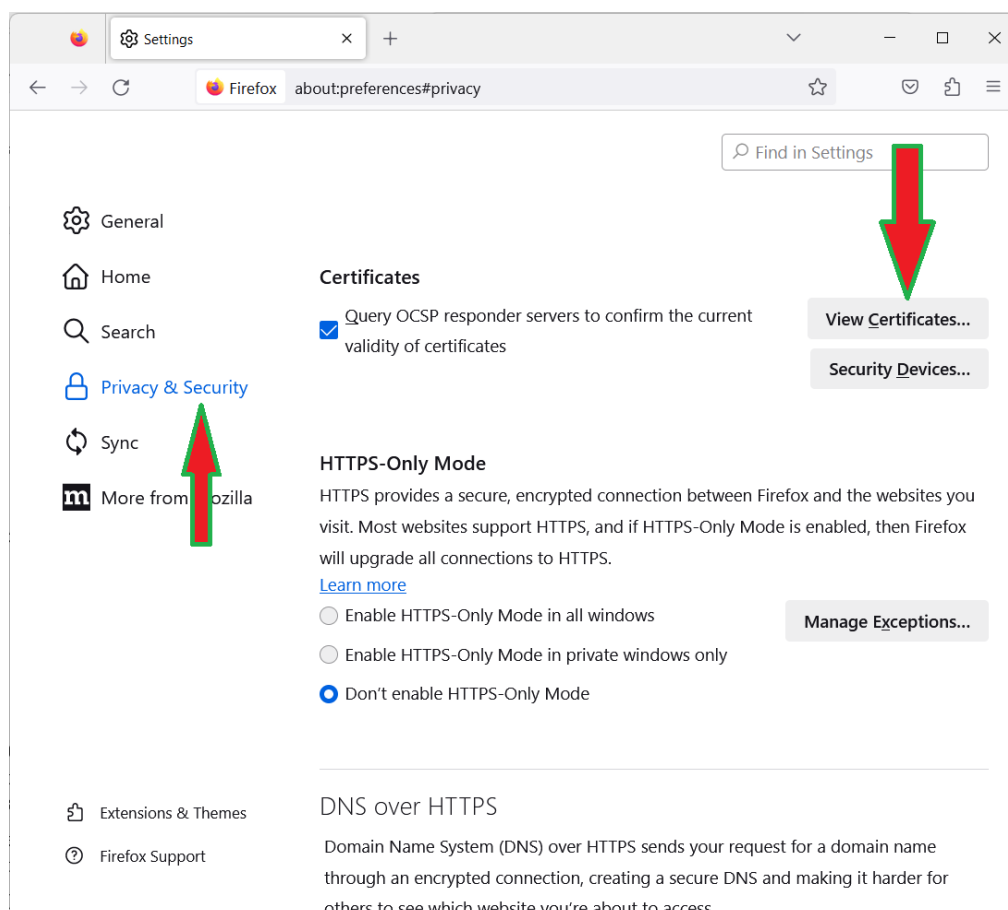
Step 06: In the **Connection Settings** window, select **Manual proxy configuration** and enter the details of proxy for HTTP and HTTPS. By default, the proxy settings configured by Burp Suite is 127.0.0.1:8080. Enter the IP address **127.0.0.1** and port number **8080** in **HTTP Proxy** and **HTTPS Proxy** textboxes.

(Please Note: In some situation, if the port number 8080 is already assigned to some other processes, then the default proxy details will be different. Hence verify the actual proxy details before entering).

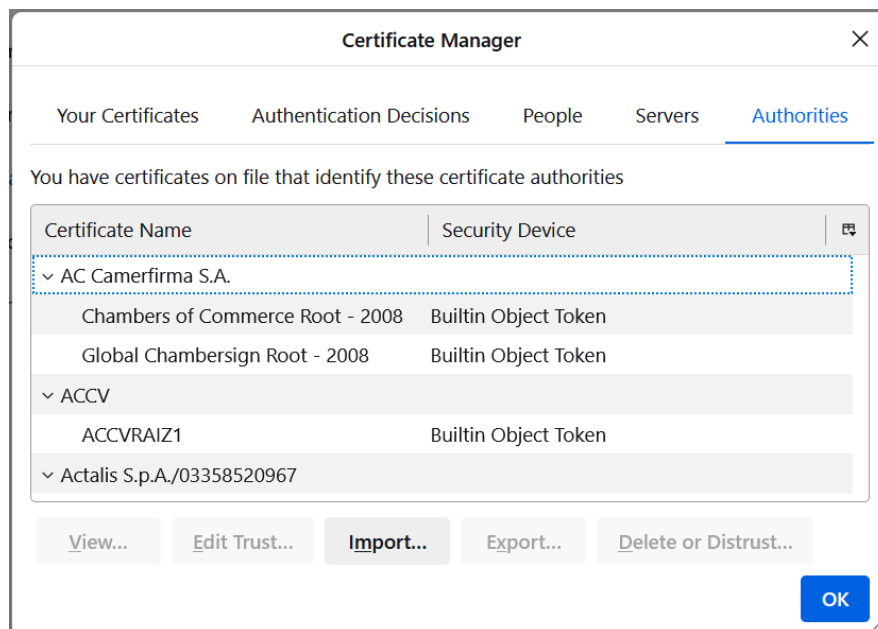
Click OK after entering the proxy details.



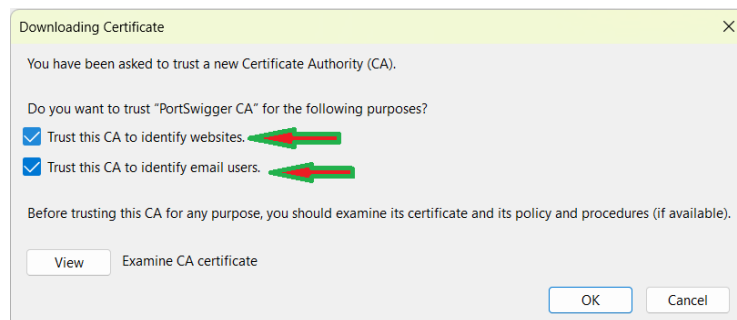
Step 07: Click on **Privacy and Security** in left panel, scroll down to **Certificates** and click on **View Certificates** button to open **Certificates Manager**.



Step 08: In the **Certificate Manager** window's **Authorities** tab, click on **Import...** button and select the file containing **"PortSwigger CA Certificate"** i.e. **cacert.der** and click **Open** button in File open dialog



Step 09: Before importing the certificate to browser, we must trust this certificate to identify websites and email users, hence select all the trust options and click **OK** button.



Step 10: Now the certificate will be imported and click **OK** button to close the **Certificate Manager** window.

Step 11: Open the Firefox and try to open any website. Now we can't reach the website because of proxy which is controlled by **Burp Suite**.

Step 12: Perform HTML Injection (GET Request):

- In Firefox, navigate to the BWAPP login page.
- Log in with bee as the username and bug as the password.
- In Burp Suite, ensure the **Intercept** is turned on.
- Select **HTML Injection (Reflected)** with **GET** method in BWAPP.
- Click **Hack** on BWAPP.
- You will see a **POST** request in Burp Suite, forward it.
- The second request is a **GET** request, and you need to modify it.
- Change the request to:

GET /htmli_get.php?firstname=prajwal+&lastname=jogi<h1>GPT</h1>&form=submit HTTP/2

- Click Forward and observe the changes on the BWAPP page:



Step 13: Perform HTML Injection (POST Request)

- Turn off the intercept mode in Burp Suite.
- Open a new tab, log in to BWAPP.
- Select HTML Injection (Post) and click Hack.
- Enter your first name and last name.
- In Burp Suite, turn on the intercept mode.
- Click Go in BWAPP.
- Modify the POST request to:
`firstname=prajwal&lastname=jogi<h1>GPT</h1>&form=submit`
- Forward the request and observe the changes on the BWAPP page.



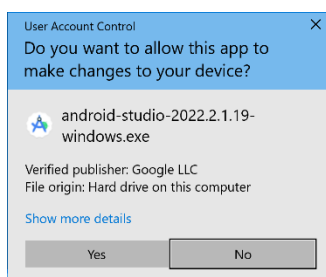
Expt. 18) Download and install the android studio and create an AVD**Downloading installing Android Studio**

Step 01: Open any browser and enter the URL <https://developer.android.com/studio> to open official website.

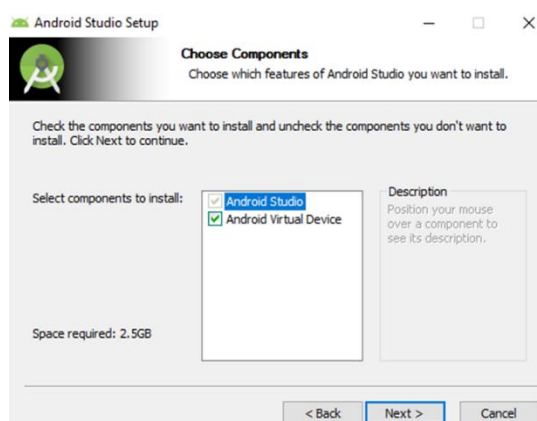
Step 02: Now click **Download Android Studio** option. Scroll down to accept **Terms and Conditions** and press **Download Android Studio**

Step 03: Wait until download complete

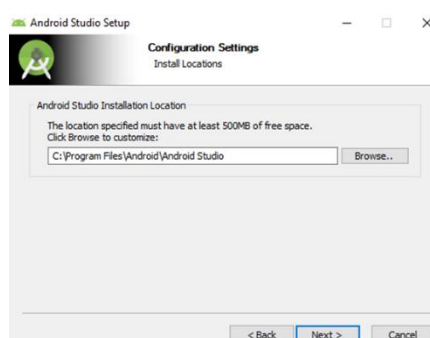
Step 04: After the download is completed, go to Downloads folder and double click it to initiate the installation process of Android Studio. Select **Yes** in **User Account Control** to continue the installation.



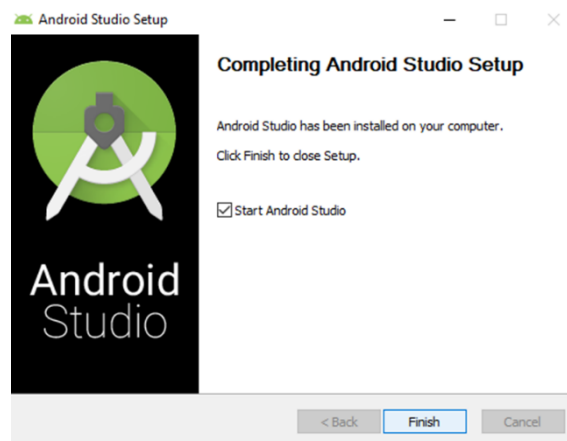
Step 05: Choose **Components** and click **Next**.



Step 06: Select **Installation Location** and click **Next**.

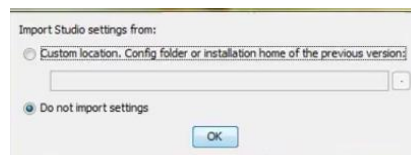


Step 07: Wait until installation complete. After the completion of installation, an option “**Start Android Studio**” will be displayed. We may uncheck this option if we don’t want to run the software immediately after the setup. Now click on Finish button to complete the process.

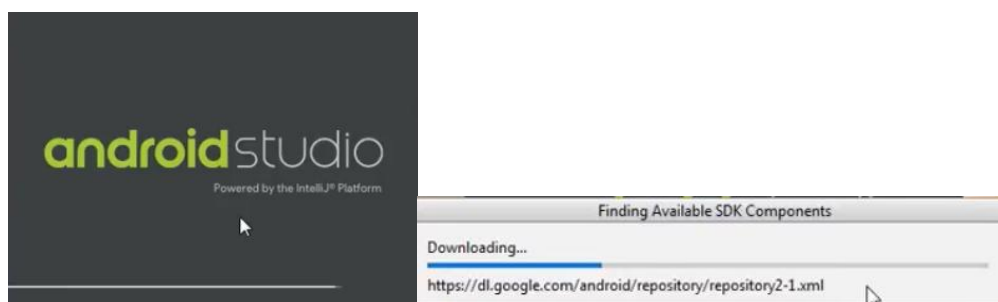


First run of Android Studio.

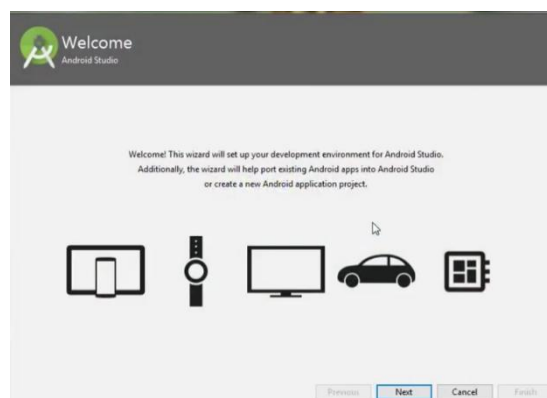
Step 01: Start the Android Studio from menu or desktop icon. It will ask whether the previous settings need to be imported [if the android studio had been installed earlier], or not. It is better to choose the "Don't import Settings option". Click the OK button.



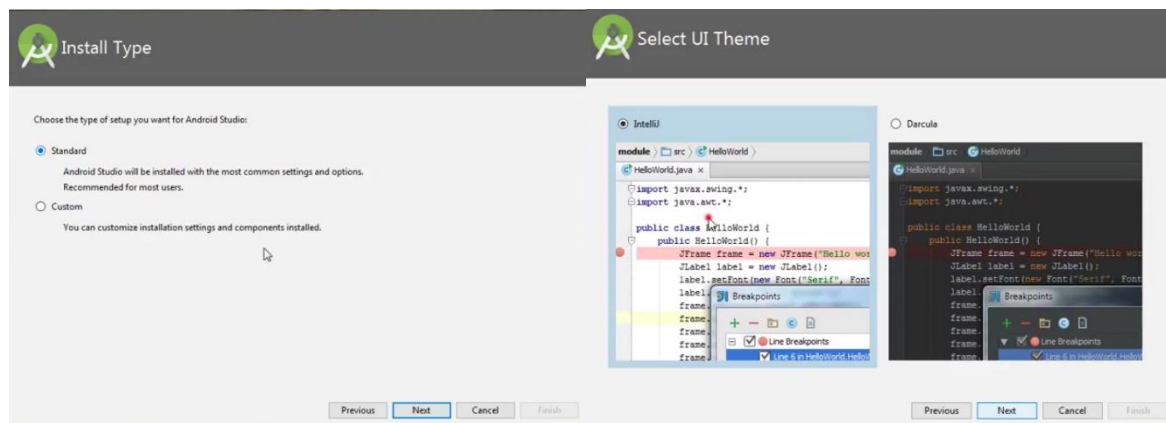
Step 02: This will start the Android Studio. Meanwhile, it will be finding the available SDK components.



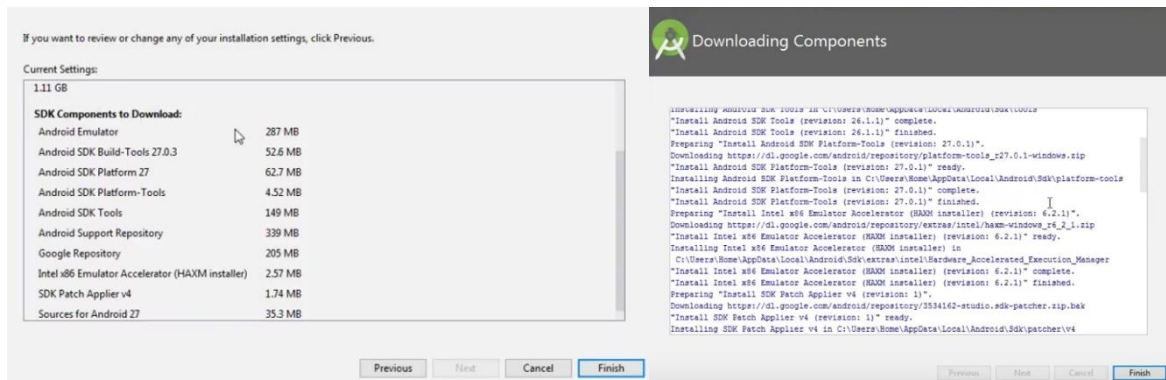
Step 03: After it has found the SDK components, it will redirect to the Welcome dialog box. Click on **Next**.



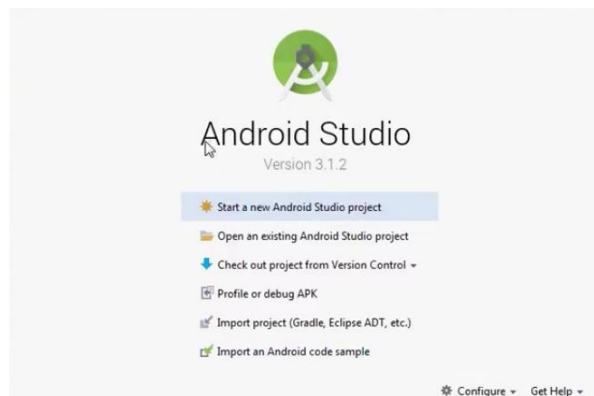
Step 04: Choose **Standard** and click on **Next**. Now choose the theme. Click on the **Next** button.



Step 05: Now it is time to download the SDK components.



Click on **Finish**. Components begin to download let it complete.



The Android Studio has been successfully configured. Now it's time to launch and build apps. Click on the Finish button to launch it.

Creating AVD in Android Studio.

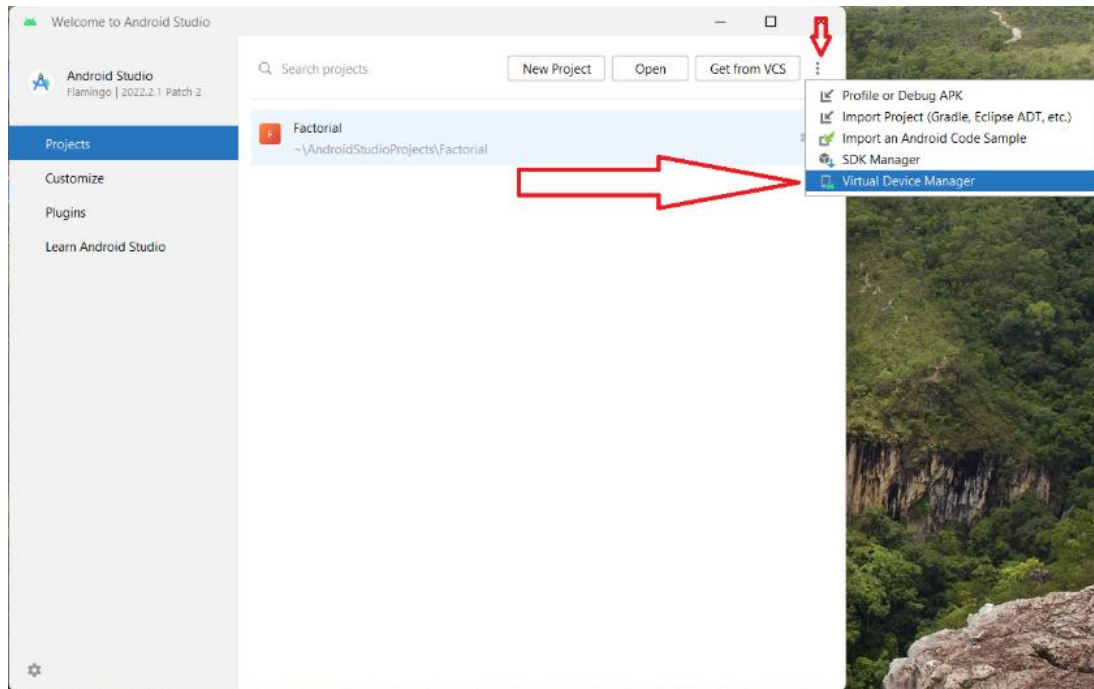
We can start **Virtual Device Manager** in two ways

- from Welcome screen of Android Studio
- from Tools menu in Android Studio IDE

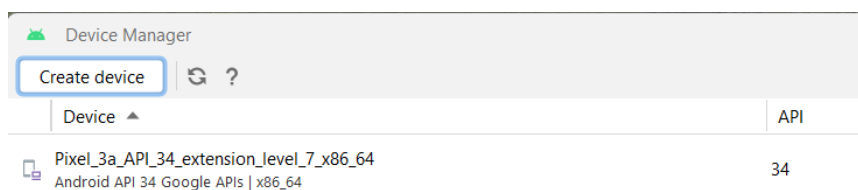
After opening Virtual Device Manager, remaining steps are same for both ways

a) From Welcome screen of Android Studio

Step 01: Start Android Studio and click on **3 dots** menu in Welcome windows and select **Virtual Device Manager**

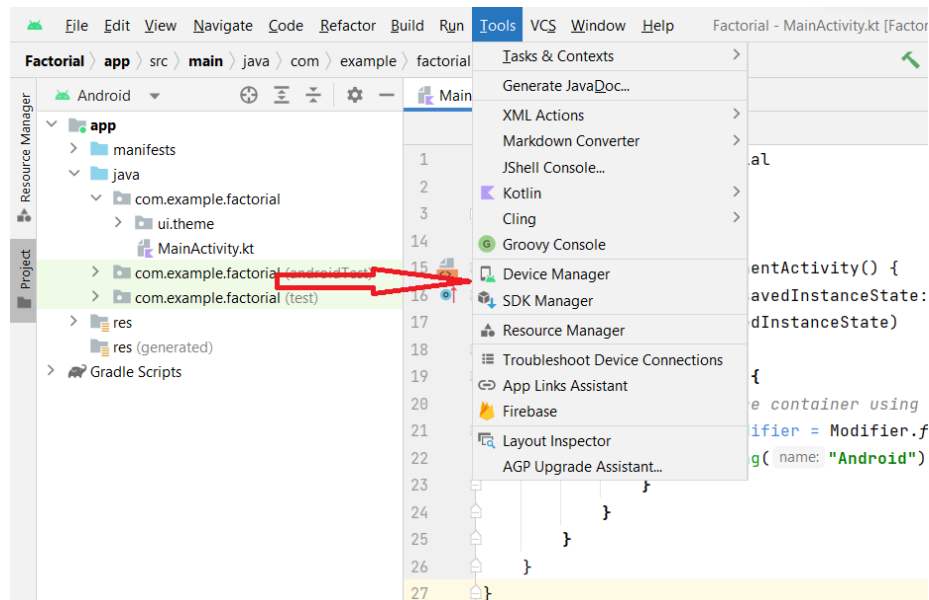


Step 02: Click on **Create device**

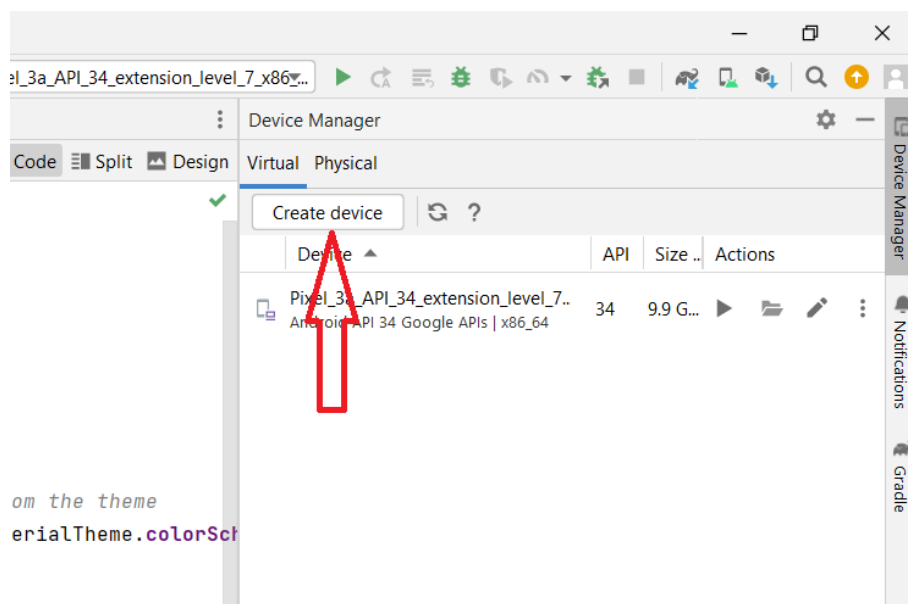


b) From Tools menu in Android Studio IDE

Step 01: Click on **Tools** menu in Android Studio IDE and select **Virtual Device Manager**

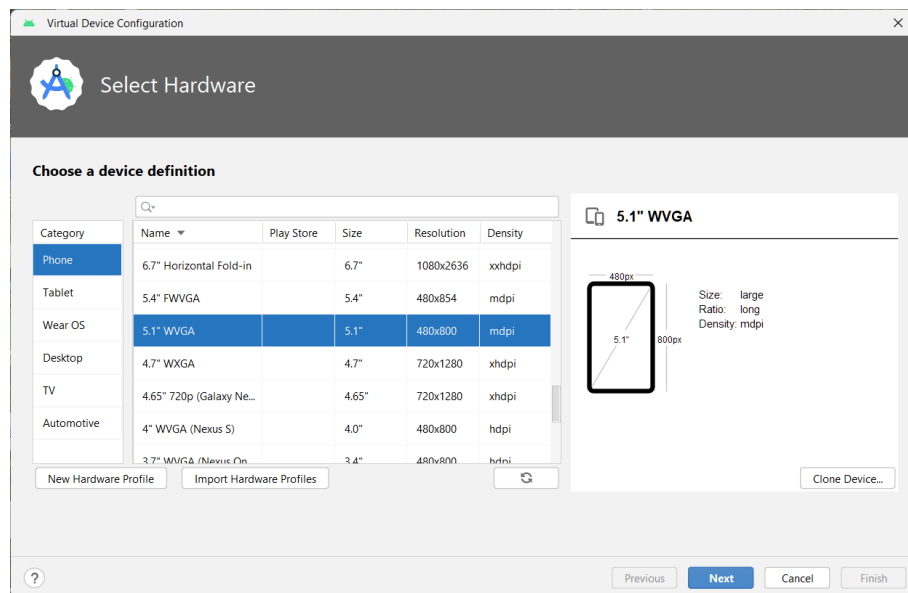


Step 02: Click on Create device

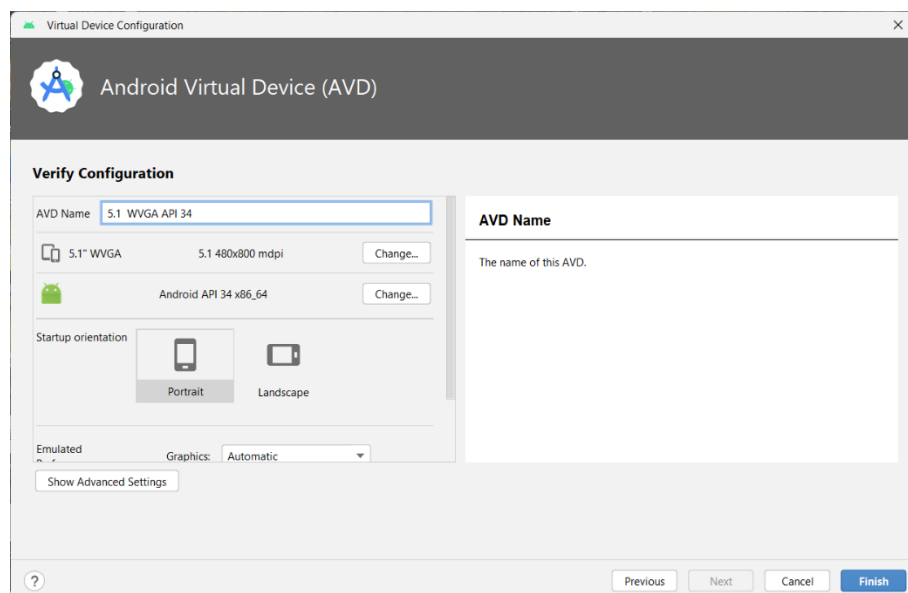


Remaining steps

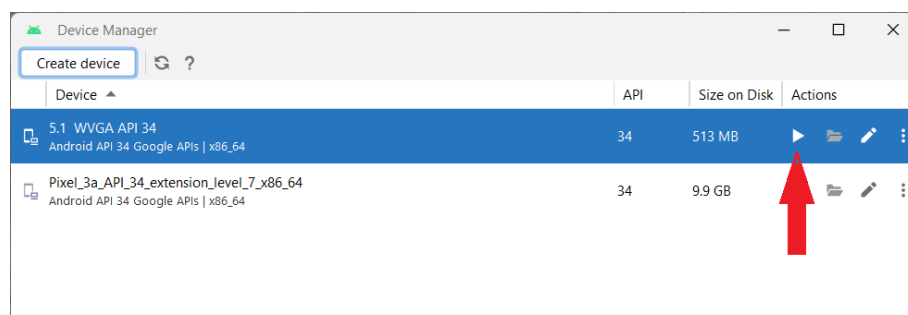
Step 03: Select the type of device definition



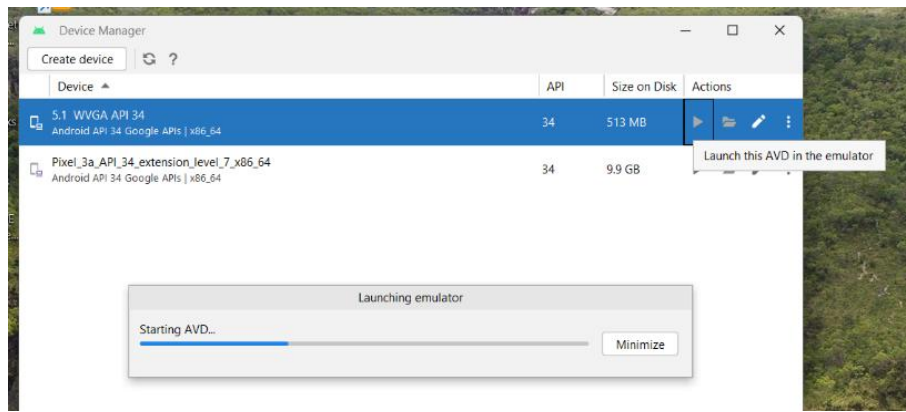
Step 04: Give a name for the new AVD and click on **Finish** button



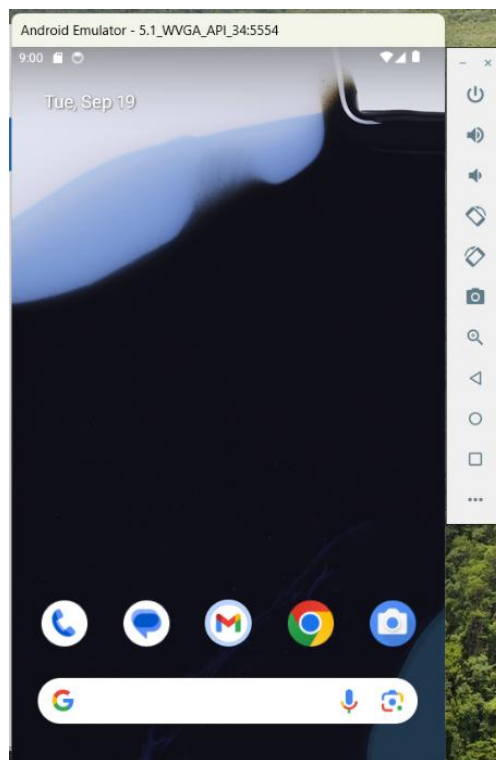
Step 05: New AVD device is just created. To start the AVD click on **run** icon under **Actions** to start the AVD



Step 06: Now the new AVD will be loaded into memory. It will take some time to completely load in to memory.

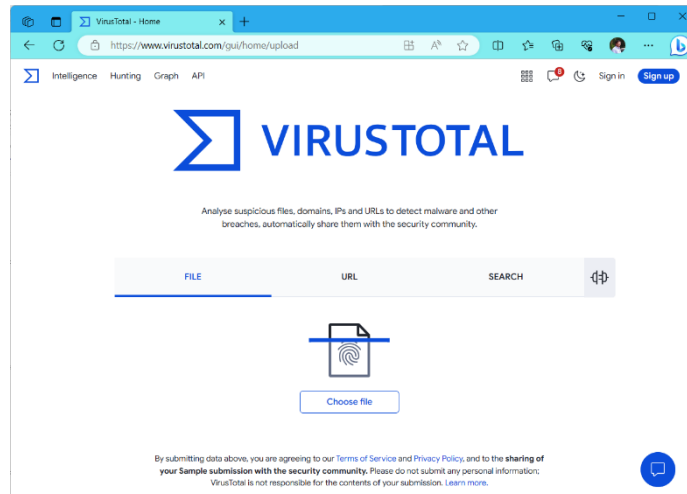


Step 07: Now the new AVD loaded successfully and we can use this virtual device for testing apps developed in Android Studio.



Expt. 19) Using VIRUSTOTAL website analyse any File, URL & Domain etc

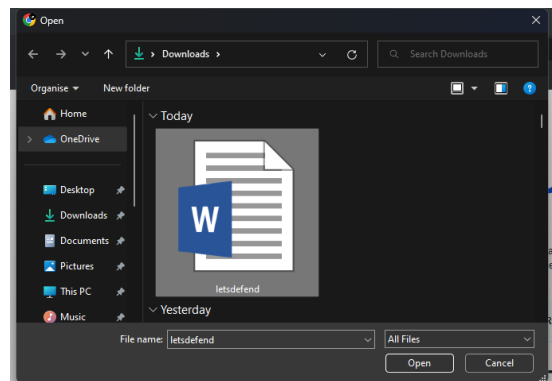
Step 01: Go to any browser and open the website <https://www.virustotal.com>



Step 02: Select **File** option and Click **Choose file** to open and scan a file



Step 03: Select any file and click **Open**



Now the website will scan the file for malwares and provides a report on it.

Step 04: Now select **URL** option and enter any URL and click enter to start a scan



Analyse suspicious files, domains, IPs and URLs to detect malware and other breaches, automatically share them with the security community.

FILE

URL

SEARCH



www.instagram.in

By submitting data above, you are agreeing to our Terms of Service and Privacy Policy, and to the sharing of your

Now the website will scan the specified URL for malwares and vulnerabilities and provides a report on it.

Step 5: Now select **SEARCH** and enter URL, Domain or IP address and press enter



Analyse suspicious files, domains, IPs and URLs to detect malware and other breaches, automatically share them with the security community.

FILE

URL

SEARCH



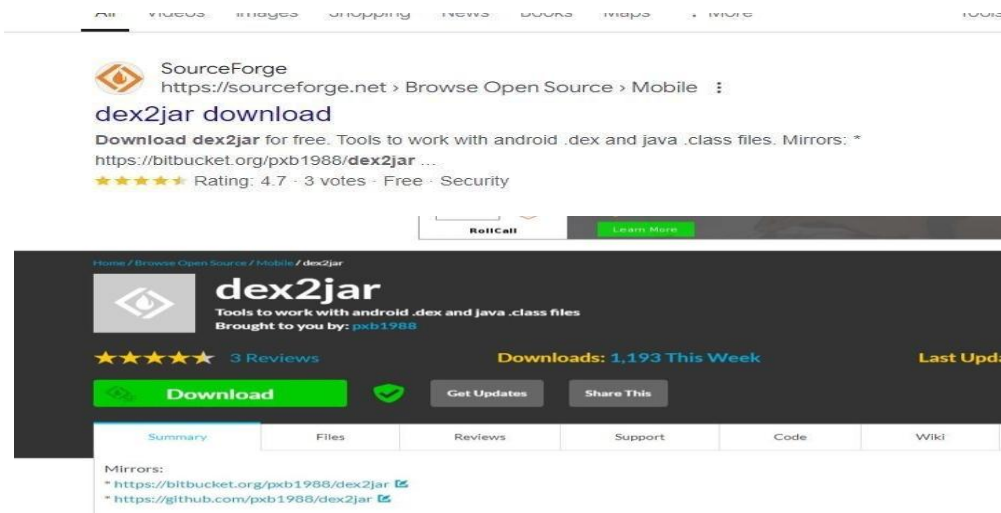
35.186.238.104

By submitting data above, you are agreeing to our Terms of Service and Privacy Policy, and to the sharing of your
Sample submission with the security community. Please do not submit any personal information; VirusTotal is
not responsible for the contents of your submission. [Learn more](#)

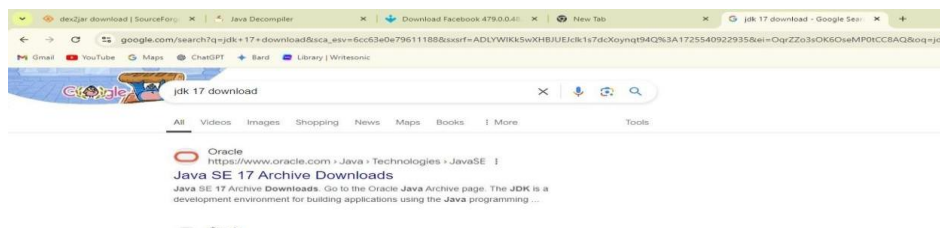
Now the website will scan the specified resources for malwares and vulnerabilities and provides a report on it.

Expt. 20) Reverse Engineering Android Application Package

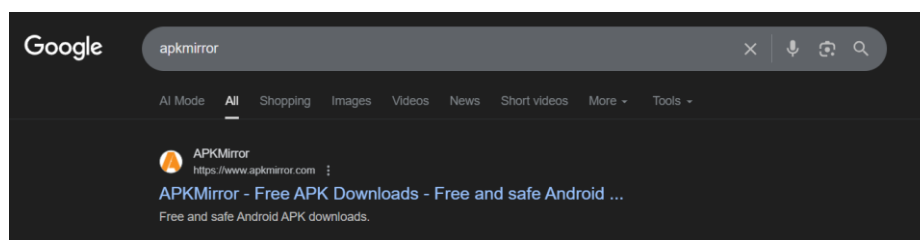
Step 1: Go to Chrome, type **dex2jar download** > navigate to SourceForge website, and then click download



Step 2: Go to Chrome, type **JDK 17 download**, and then go to Oracle website, and then click the windows x64 Installer



Step 3: Go to Chrome, type **Facebook apk download** or any application apk file, and then go to apkmirror website and then click download

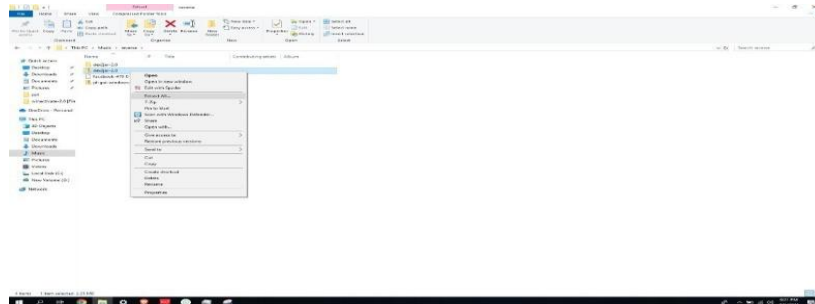


Step 4: Go to Chrome, type **jd-gui download**, and then go to java decompiler website then click the Windows icon and download it

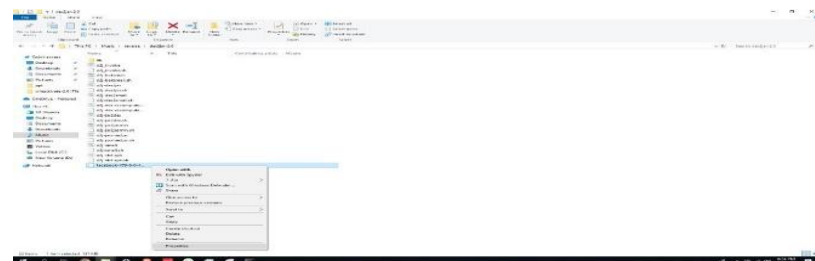




Step 5: Then extract the dex2jar folder and jd gui folder

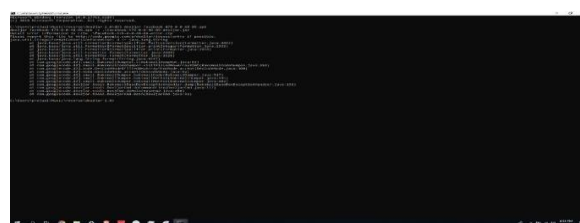
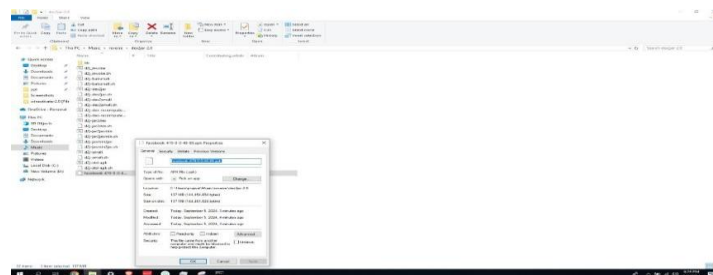


Step 6: Then copy the downloaded apk file and paste it inside of dex2jar folder

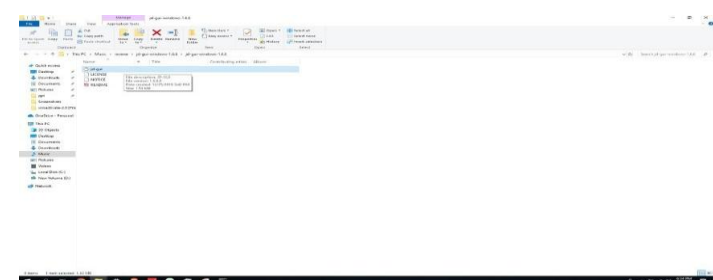


Step7: In the dex2jar folder's address bar, type **cmd** and press Enter.

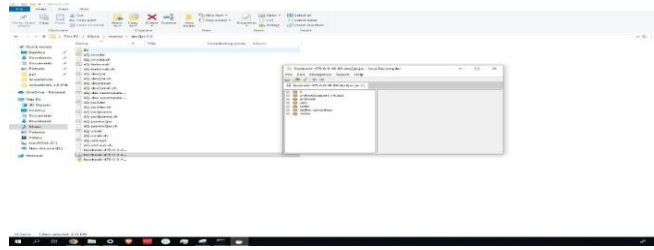
Step 8: Copy the path of apk file and type the command in cmd> **d2j-dex2jar** (then paste the path of the apk file)



Step 9: Then open the JD gui folder then run the JD gui application and minimize it



Step 10: Then go to dex2jar folder then drag and drop the Facebook apk jar file into jd-gui application



Then we will see the source code apk file

Expt. 21) Demonstrate the use of NMAP tool.**TCP Connect Scan (-sT)**

- A TCP Connect scan determines if a port is open by completing the full three-way TCP handshake (SYN, SYN/ACK, ACK) with the target.
- **Syntax:** nmap -sT <target_ip>

```
Starting Nmap 7.94 ( https://nmap.org ) at 2025-09-22 10:31 IST
Nmap scan report for 192.168.1.10 (Windows 11 Machine)
Host is up (0.0021s latency).
Not showing: 996 closed tcp ports
PORT      STATE      SERVICE
135/tcp    open       msrpc
139/tcp    open       netbios-ssn
445/tcp    open       microsoft-ds
5040/tcp   filtered   unknown
Nmap done: 1 IP address (1 host up) scanned in 0.58 seconds
```

SYN "Stealth" Scan (-sS)

- A SYN scan, often called a "stealth" or "half-open" scan, sends only a TCP SYN packet and determines the port's state from the response (SYN/ACK or RST) without completing the connection, making it less likely to be logged.
- **Syntax:** sudo nmap -sS <target_ip> (Requires administrator/root privileges)

```
Starting Nmap 7.94 ( https://nmap.org ) at 2025-09-22 10:31 IST
Nmap scan report for 192.168.1.10 (Windows 11 Machine)
Host is up (0.0019s latency).
Not showing: 996 filtered tcp ports
PORT      STATE      SERVICE
135/tcp    open       msrpc
139/tcp    open       netbios-ssn
445/tcp    open       microsoft-ds
3389/tcp   closed     ms-wbt-server
Nmap done: 1 IP address (1 host up) scanned in 4.21 seconds
```

ACK Scan (-sA)

- An ACK scan sends TCP ACK packets to probe firewall rulesets and determine if a port is **filtered** or **unfiltered**, but it cannot determine if a port is actually open.
- **Syntax:** sudo nmap -sA <target_ip> (Requires administrator/root privileges)

```
Starting Nmap 7.94 ( https://nmap.org ) at 2025-09-22 10:31 IST
Nmap scan report for 192.168.1.10 (Windows 11 Machine)
Host is up (0.0025s latency).
All 1000 scanned ports on 192.168.1.10 are in ignored states.
Not showing: 997 filtered ports
PORT      STATE      SERVICE
80/tcp     unfiltered  http
135/tcp     unfiltered  msrpc
443/tcp     unfiltered  https
Nmap done: 1 IP address (1 host up) scanned in 3.15 seconds
```

UDP Scan (-sU)

- A UDP scan sends protocol-specific UDP packets to determine if a port is open, but it is often slow and reports ports as open|filtered due to the connectionless nature of UDP.
- **Syntax:** `sudo nmap -sU <target_ip>` (Requires administrator/root privileges)

```
Starting Nmap 7.94 ( https://nmap.org ) at 2025-09-22 10:31 IST
Nmap scan report for 192.168.1.10 (Windows 11 Machine)
Host is up (0.0015s latency).
Not showing: 997 closed udp ports
PORT      STATE      SERVICE
137/udp    open|filtered netbios-ns
500/udp    open|filtered isakmp
5353/udp   open|filtered mdns

Nmap done: 1 IP address (1 host up) scanned in 1802.43 seconds
```

Full / Aggressive Scan (-A)

- An "aggressive" or comprehensive scan combines multiple techniques, including OS detection (-O), service version detection (-sV), script scanning (-sC), and traceroute (--traceroute), to gather as much information as possible about the target.
- **Syntax:** `sudo nmap -A <target_ip>` (Requires administrator/root privileges)

```
Starting Nmap 7.94 ( https://nmap.org ) at 2025-09-22 10:31 IST
Nmap scan report for 192.168.1.10
Host is up (0.0031s latency).
Not showing: 997 closed tcp ports
PORT      STATE SERVICE      VERSION
135/tcp    open  msrpc        Microsoft Windows RPC
139/tcp    open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds Windows 11 microsoft-ds
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
|_smb2-time: OS: Windows 11 (build 22621)
|_smb-os-discovery: OS: Windows 11 (Windows 11)

Device type: general purpose
Running: Microsoft Windows 11
OS CPE: cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 11

Nmap done: 1 IP address (1 host up) scanned in 15.34 seconds
```

NULL Scan (-sN)

- A NULL scan sends a TCP packet with no flags set, which should elicit an RST response from a closed port on a compliant system, but this method is **ineffective against Windows**.
- **Syntax:** `sudo nmap -sN <target_ip>` (Requires administrator/root privileges)

```
Starting Nmap 7.94 ( https://nmap.org ) at 2025-09-22 10:31 IST
Nmap scan report for 192.168.1.10 (Windows 11 Machine)
Host is up (0.0018s latency).
All 1000 scanned ports on 192.168.1.10 are in state open|filtered

Nmap done: 1 IP address (1 host up) scanned in 20.11 seconds
```